



Massachusetts Institute of Technology

2.C51

Modeling of a Roll-to-Roll Continuous Manufacturing Process

Merlin Gogolin, Mia Galatis, Samuel Gomez, Sankaran Iyer,
Sebastian Podiono

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What is Roll-to-Roll Manufacturing?

Inputs

- Packaging Film
- Liquid Consumable



Process



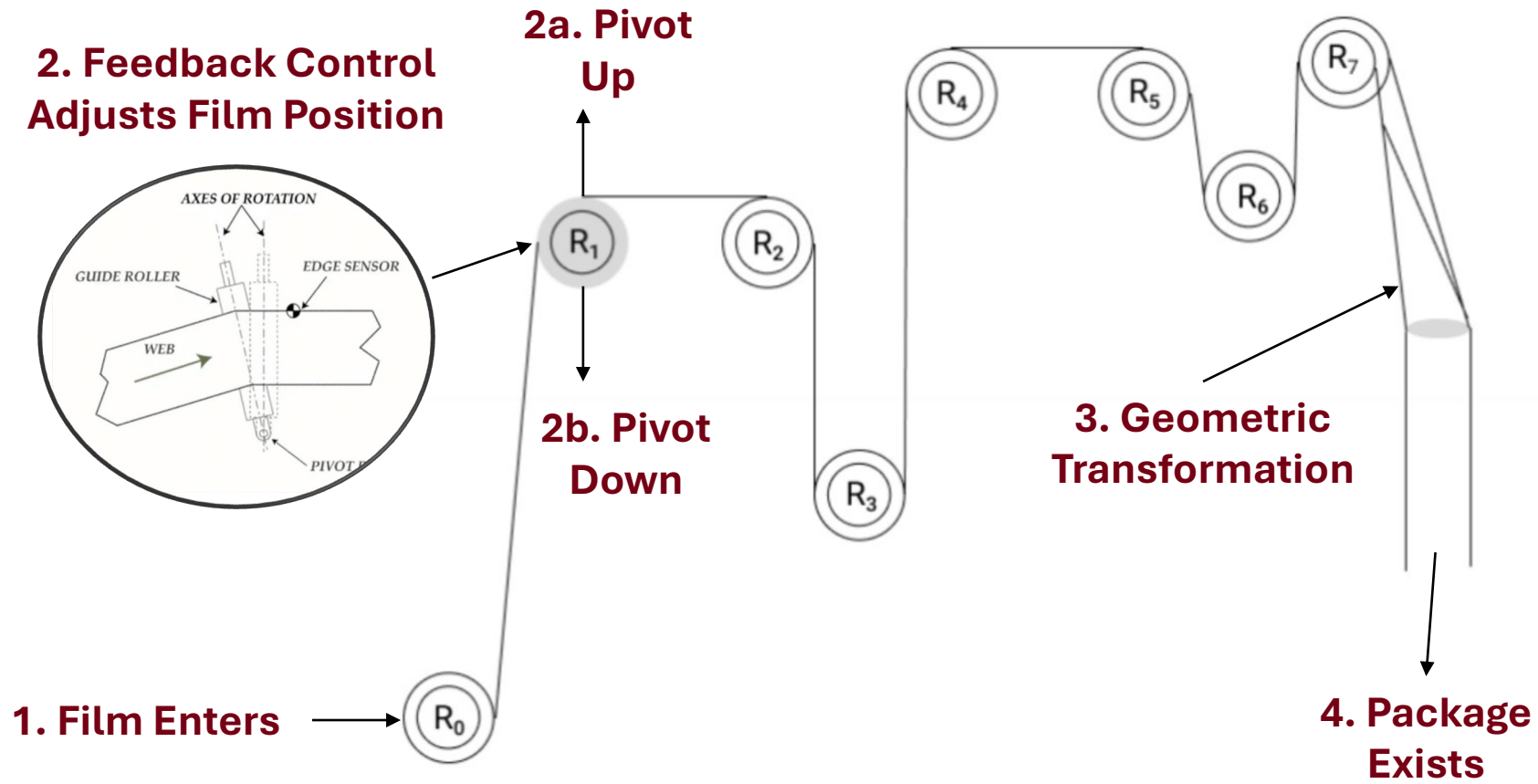
Output

- Beverage of Choice



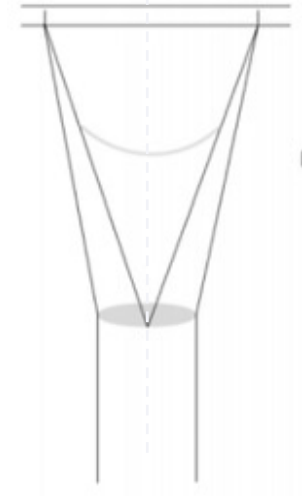
More Are About the Process

Side View

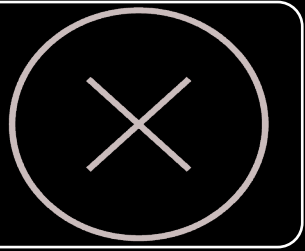


Front View

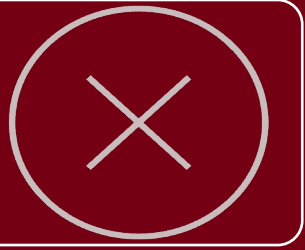
2. Feedback Setpoint – Process Value



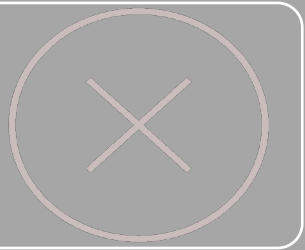
Production Opportunities



Disturbances are causing downtime and waste.



High product quality variation.



Engineer-dependent performance.

The Value Proposition

Eliminate

Eliminate
downtime and
waste related to
process
disturbances.

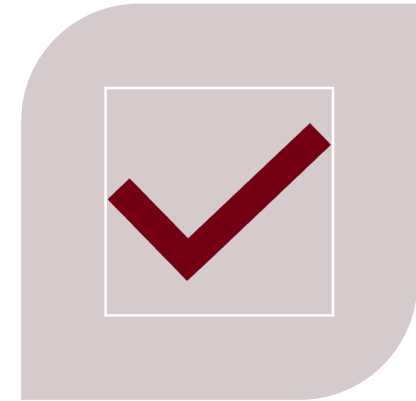
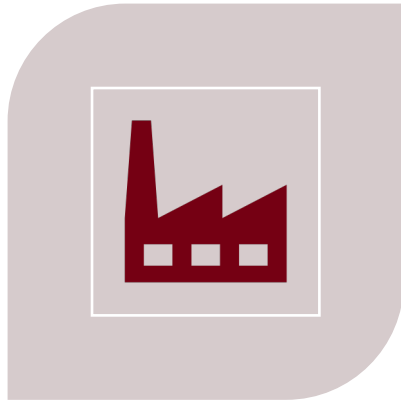
Reduce

Reduce product
variability.

Ensure Optimality

Ensure optimal,
scalable control.

Workflow



BUILD A MODEL

OPTIMIZE
CONTROLLER

DEPLOY AND
TEST

Build a Model

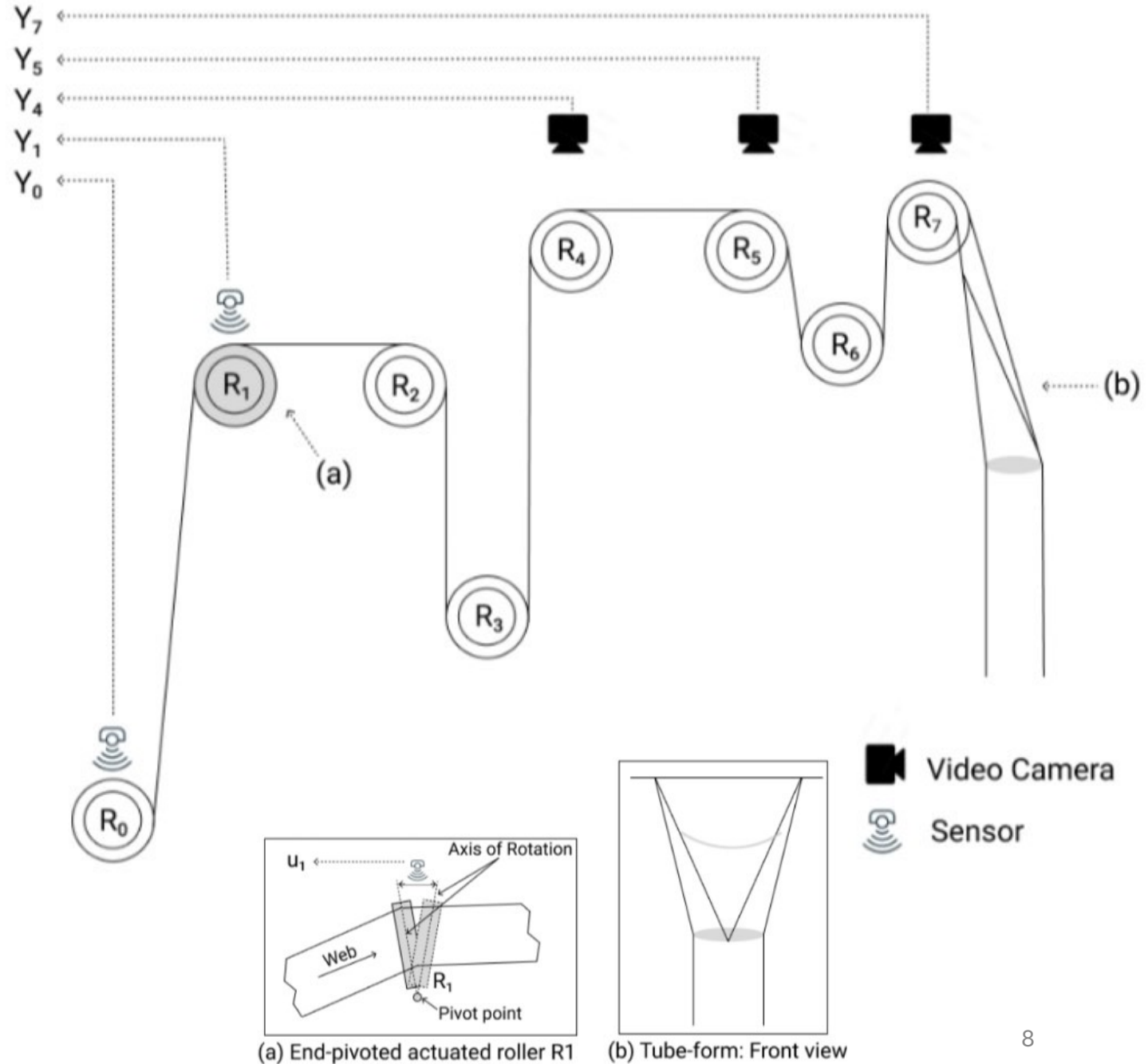
Modeling Methodology

(4) Models

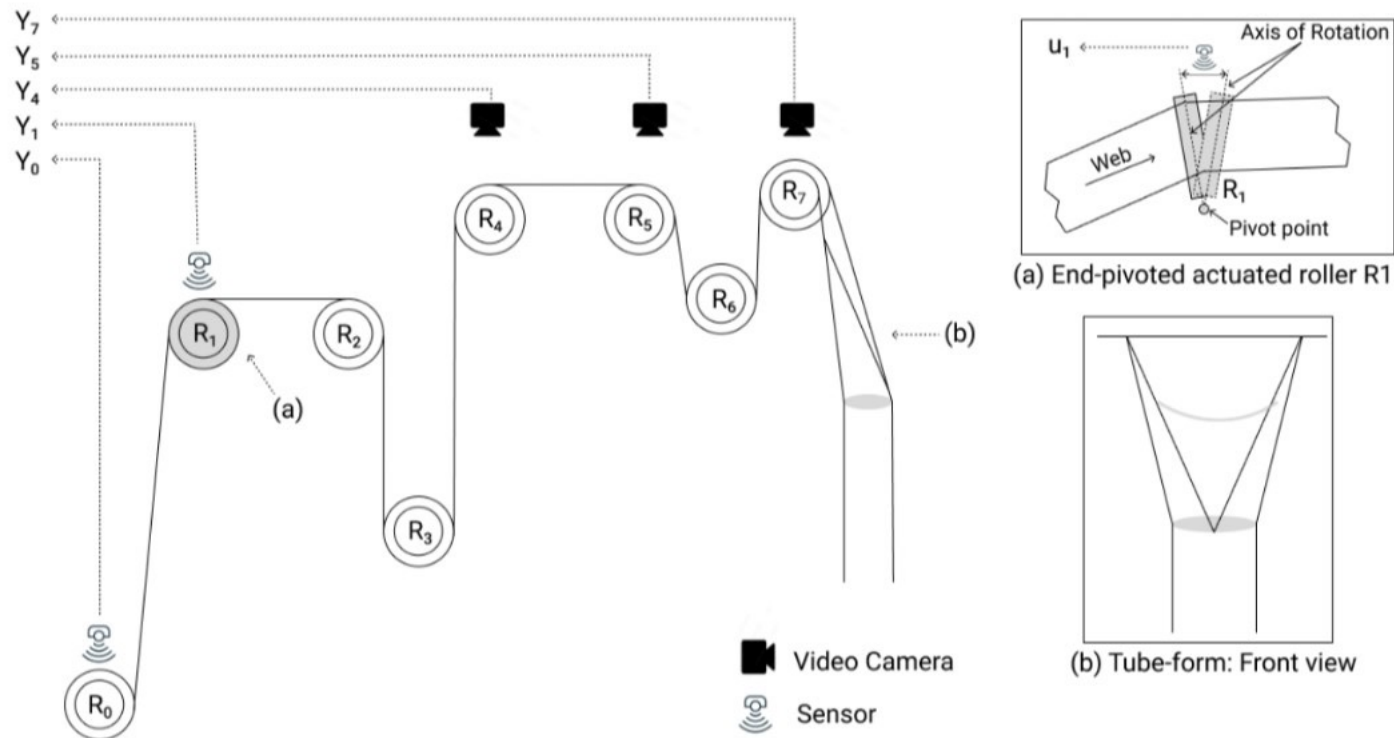
- R0-R1
- R1-R4
- R4-R5
- R5-R7

(7) Data Points

- Film Position:
 Y_0 , Y_1 , Y_4 , Y_5 , Y_7
- Actuator Position
- Film Velocity



(10) Experiments



10 Experiments

(10) Narrow-Film Experiments

21_4mm_test2, 22_2mm_fast,
23_splice, 24_step_gain_1_75,
25_step_gain_2_6, 26_splice_fast,
27_4mm_fast, 28_4mm,
29_2mm_test2, 30_2mm

Modeling

Physics-Informed Transfer Functions

Physics-Informed Modeling

The web span is modeled using Timoshenko beam under tension. The 4th-order differential equation below represents the lateral deflection of the web along the roller span:

$$\frac{d^4 y}{dx^4} - K^2 \frac{d^2 y}{dx^2} = 0, \quad K^2 = \frac{T}{EI \left(1 + \frac{T}{GA}\right)}.$$

The general solution of the lateral position $y(x)$ and web angle $\theta(x)$ are:

$$y(x) = C_1 + C_2 x + C_3 \cosh(K_\gamma x) + C_4 \sinh(K_\gamma x)$$

$$\theta(x) = C_2 + C_3 K_\gamma \sinh(K_\gamma x) + C_4 K_\gamma \cosh(K_\gamma x)$$

Here, C_1 , C_2 , C_3 , and C_4 are constants of integration that must be solved for using boundary conditions.

Physics-Informed Modeling

Solving for the four constants requires four boundary conditions.

Using Upstream Position,
Upstream Angle,
Downstream Bending Moment,
and Downstream Shear yields:

$$\mathbf{M} \begin{bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{bmatrix} = \begin{bmatrix} y_0 \\ \theta_0 \\ \text{BC}_3 \\ \text{BC}_4 \end{bmatrix}$$

Inverting the matrix to find the constants results in the following equation which characterizes how the web behaves in the span $x = L$:

$$y(L) = f_1(KL) \cdot y_0 + L \cdot f_2(KL) \cdot \theta_0 + f_3(KL) \cdot (\text{roller offsets})$$

$$f_1(KL) = \frac{(KL)^2(\cosh(KL) - 1)}{KL \sinh(KL) + 2(1 - \cosh(KL))} \quad f_2(KL) = \frac{KL(KL \cosh(KL) - \sinh(KL))}{KL \sinh(KL) + 2(1 - \cosh(KL))}$$

$$f_3(KL) = \frac{KL(\sinh(KL) - KL)}{KL \sinh(KL) + 2(1 - \cosh(KL))}$$

Physics-Informed Modeling

Substituting the static equation into the dynamics of the roller span yields:

$$L^2 y''(L) = -f_1 y(L) + f_2 L \theta(L) + f_1 y_0 + f_3 L \theta_0$$

Solving for the transfer function yields:

$$\frac{Y_j(s)}{Y_i(s)} = \frac{-\frac{f_3}{\tau_j} s + \frac{f_1}{\tau_j^2}}{s^2 + \frac{f_2}{\tau_j} s + \frac{f_1}{\tau_j^2}}$$

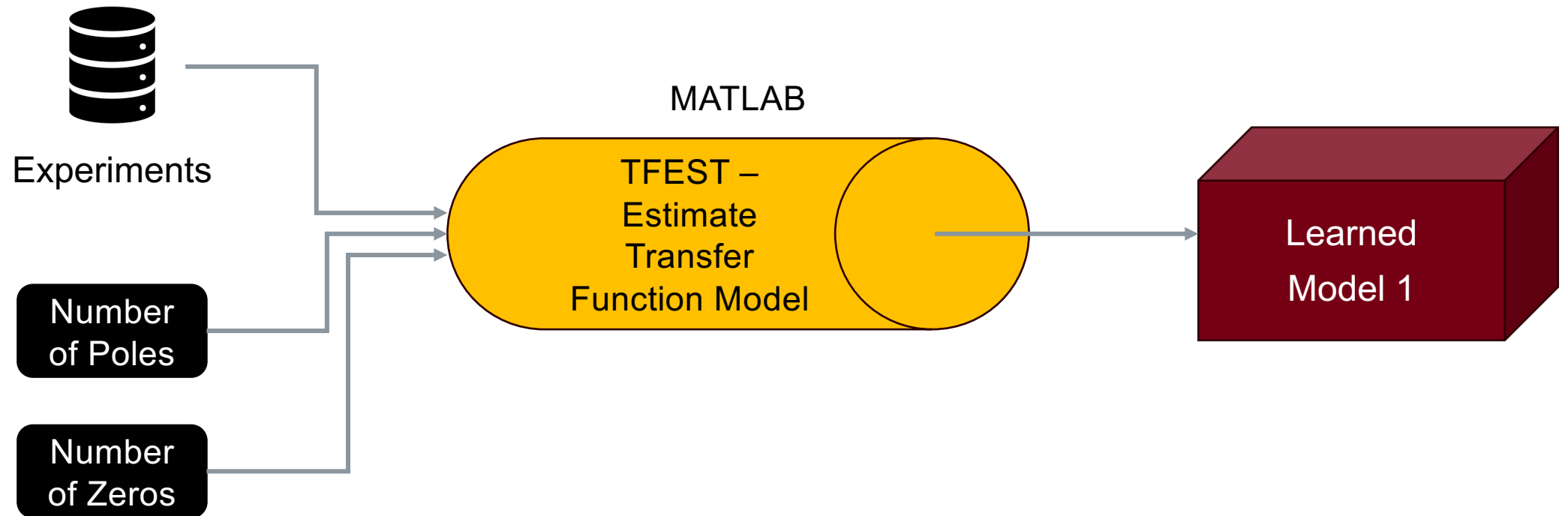
Physics-Informed Modeling

Parameters used:

Symbol	Name	Description	In MATLAB code
T	Web tension	Longitudinal pulling force on the web (N)	<code>T_ = 100</code>
EI	Bending stiffness	Product of Young's modulus E and moment of inertia I ; resists lateral curvature (N·m ²)	<code>E_ = 3.5e9 , I_ = 1/2 * b_^3 * h_</code>
GA	Shear stiffness	Product of shear modulus G and cross-sectional area A ; accounts for shear deformation (N)	<code>G_ = E_/(2*(1+mu_)) , A_ = b_ * h_</code>
v	Web speed	Transport velocity (m/s)	<code>v_ = 1.125</code>
L	Span length	Free length of web between two rollers (m)	<code>L1 , L2 , ... L7</code>
b	Web width	Cross-web dimension (m)	<code>b_ = 0.166</code>
h	Web thickness	Through-thickness dimension (m)	<code>h_ = 0.20/1000</code>

Learned Linear Transfer Functions

Learned Linear Models Informed by Physics

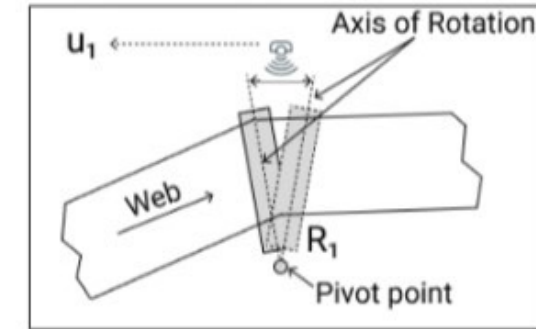
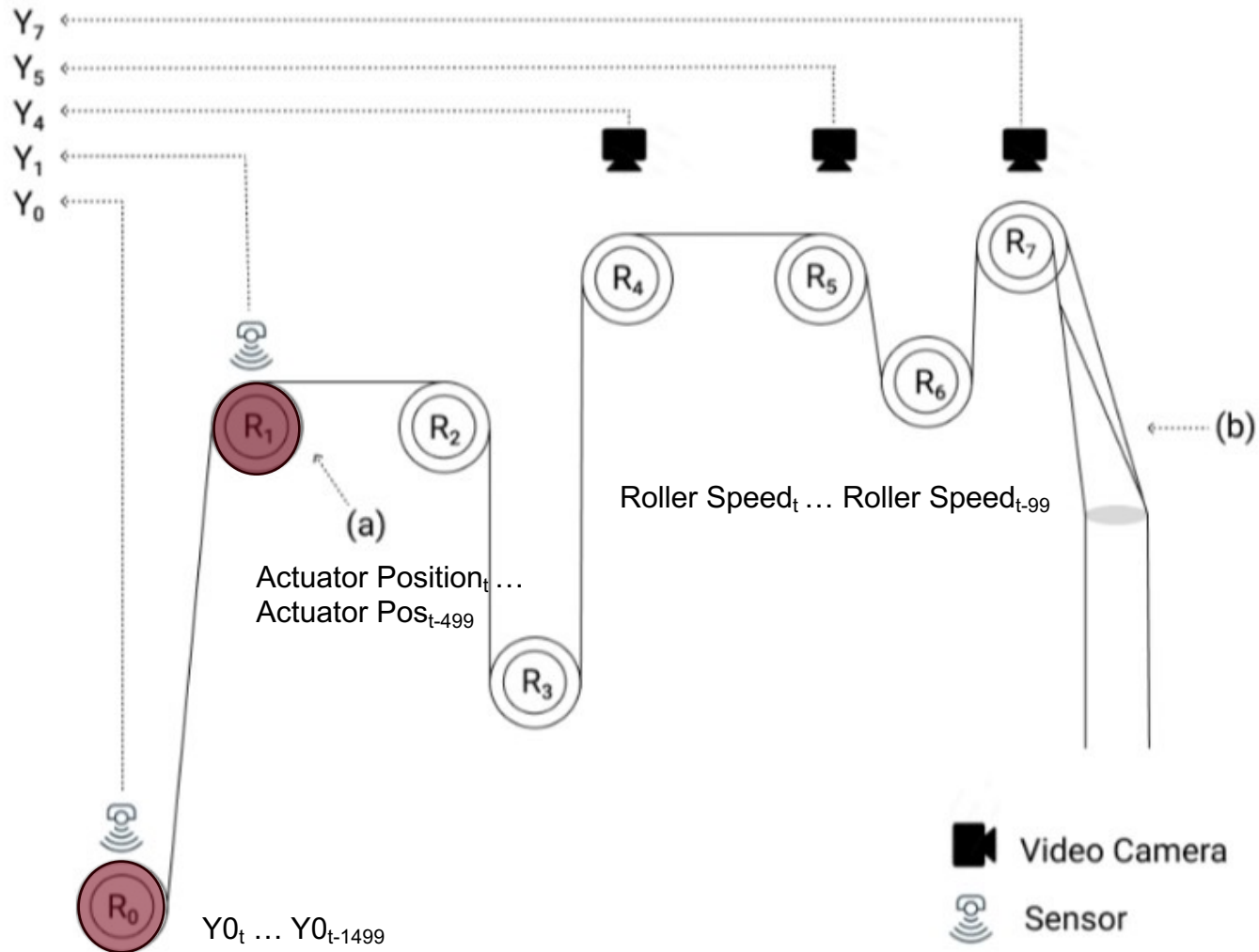


Fully-Connected Neural Network with Lagged Variables

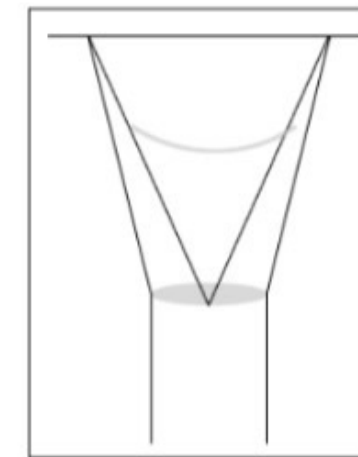
NFIR

Nonlinear Finite Impulse Response

NFIR Modeling



(a) End-pivoted actuated roller R1

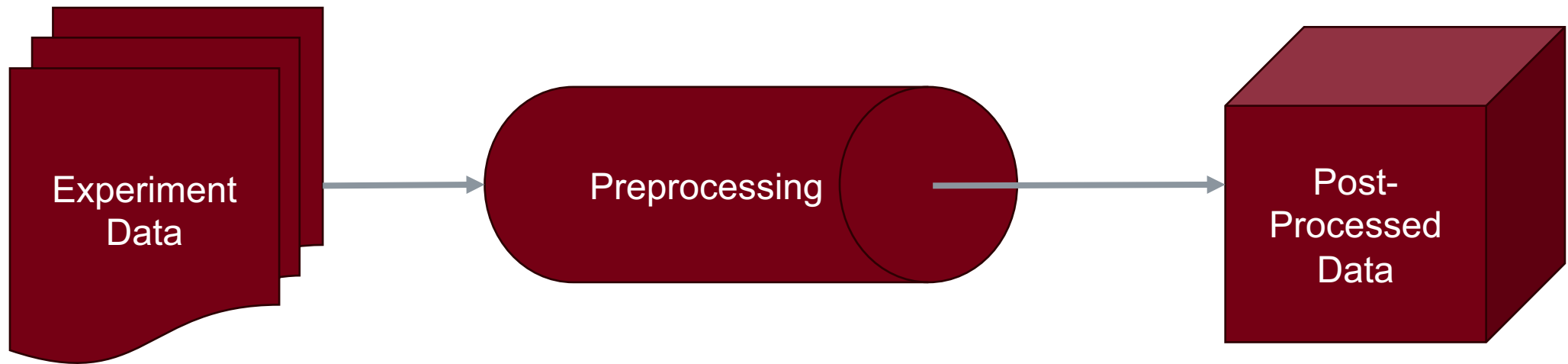


(b) Tube-form: Front view

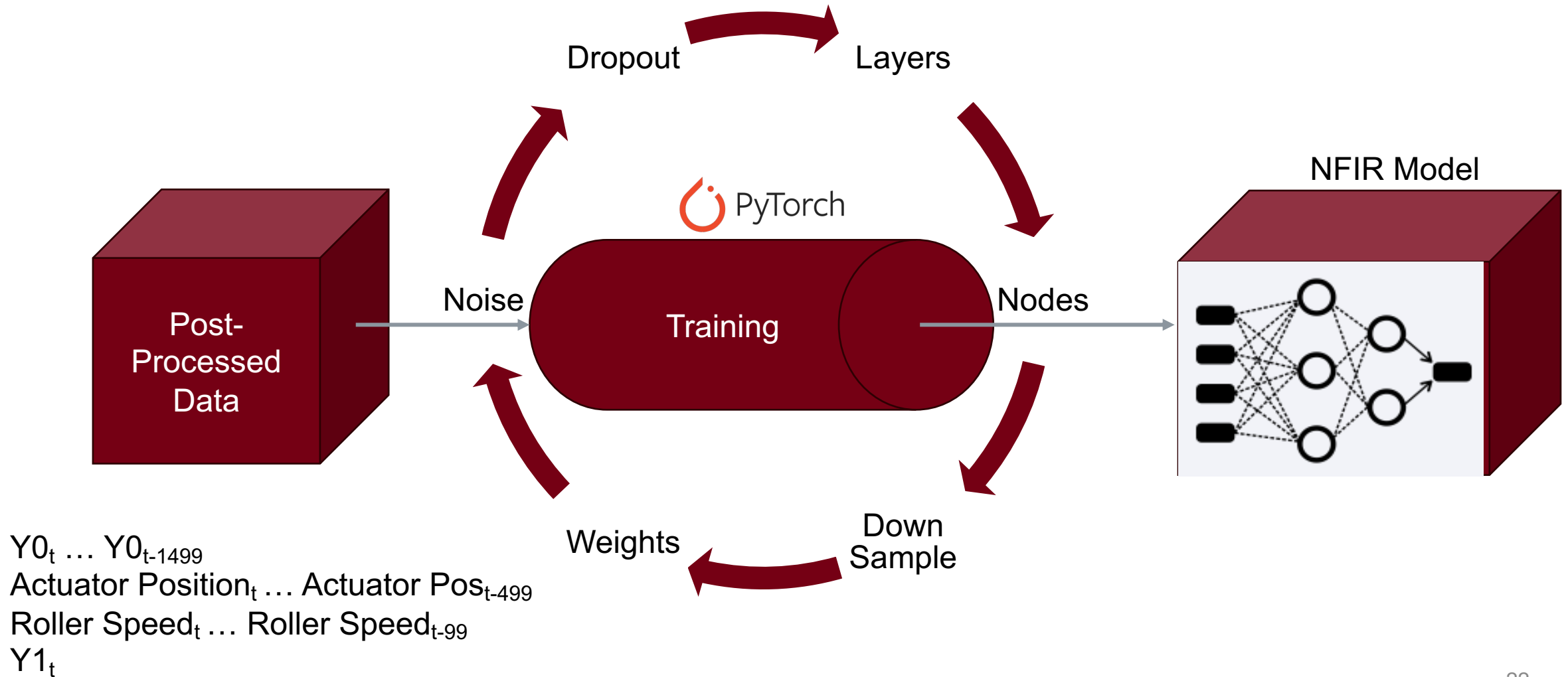
Step 1: Build a Network

```
124
125 class FullyConnectedNetwork_varDepth(nn.Module):
126     def __init__(self, input_size, hidden_size, output_size, num_layers, p_dropout):
127         super(FullyConnectedNetwork_varDepth, self).__init__()
128
129         self.hidden_layers = nn.ModuleList()
130
131         # Input layer
132         self.hidden_layers.append(weight_norm(nn.Linear(input_size, hidden_size)))
133         self.hidden_layers.append(nn.Dropout(0.2))
134
135         # Hidden layers
136         for i in range(num_layers-1):
137             next_hidden_size = max(hidden_size // (1**(i+1)), 1)
138             self.hidden_layers.append(weight_norm(nn.Linear(hidden_size, next_hidden_size)))
139             self.hidden_layers.append(nn.Dropout(p_dropout))
140             hidden_size = next_hidden_size
141
142         # Output layer
143         self.fc_out = weight_norm(nn.Linear(hidden_size, output_size))
144
145         # Activation
146         self.relu = nn.ReLU()
147
148     def forward(self, x):
149         for layer in self.hidden_layers:
150             if isinstance(layer, nn.Dropout):
151                 x = layer(x)
152             else:
153                 x = self.relu(layer(x))
154
155         x = self.fc_out(x)
156
157         return x
```

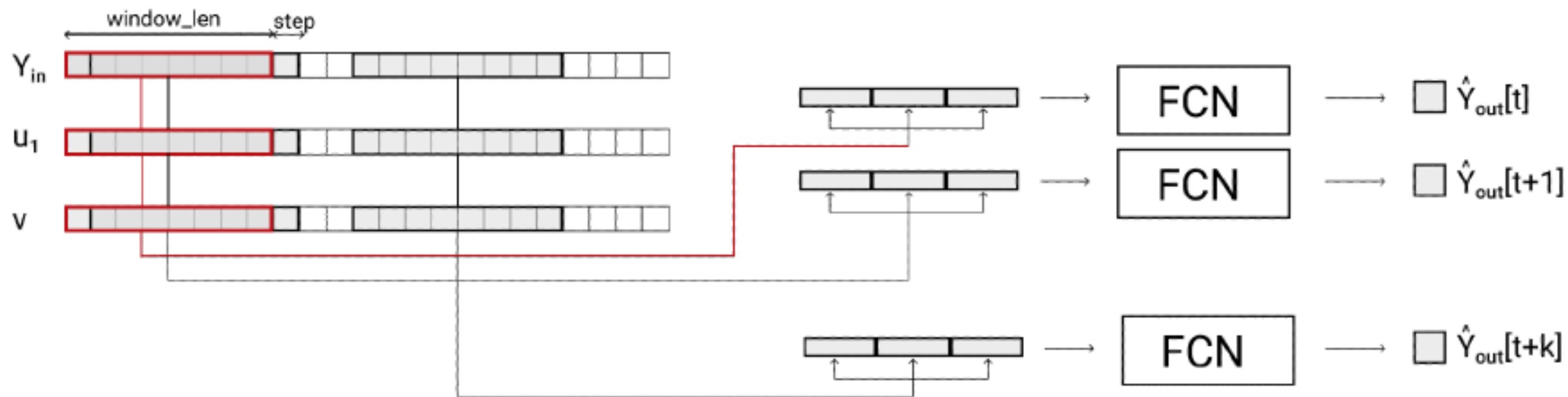
Step 2: Import and Preprocess Data



Hyperparameter Tuning and Training

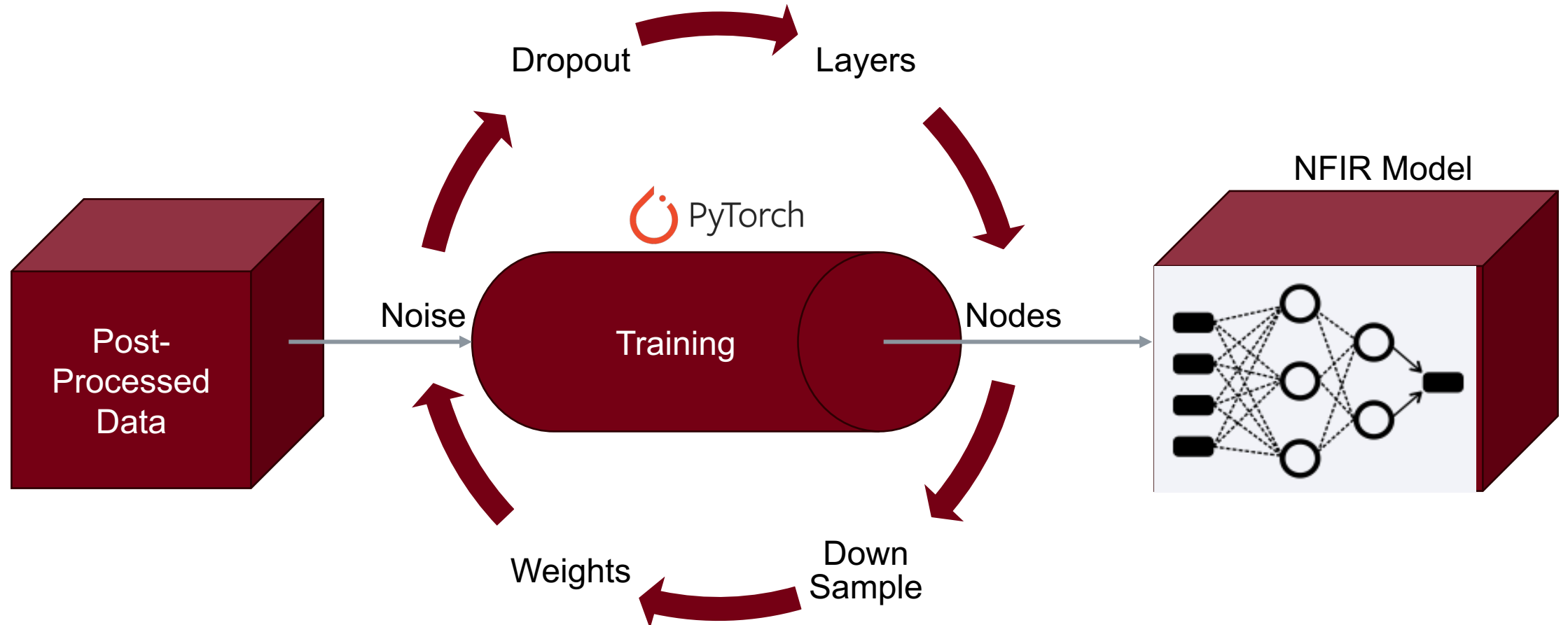


Step 3: Hyperparameter Tuning and Training



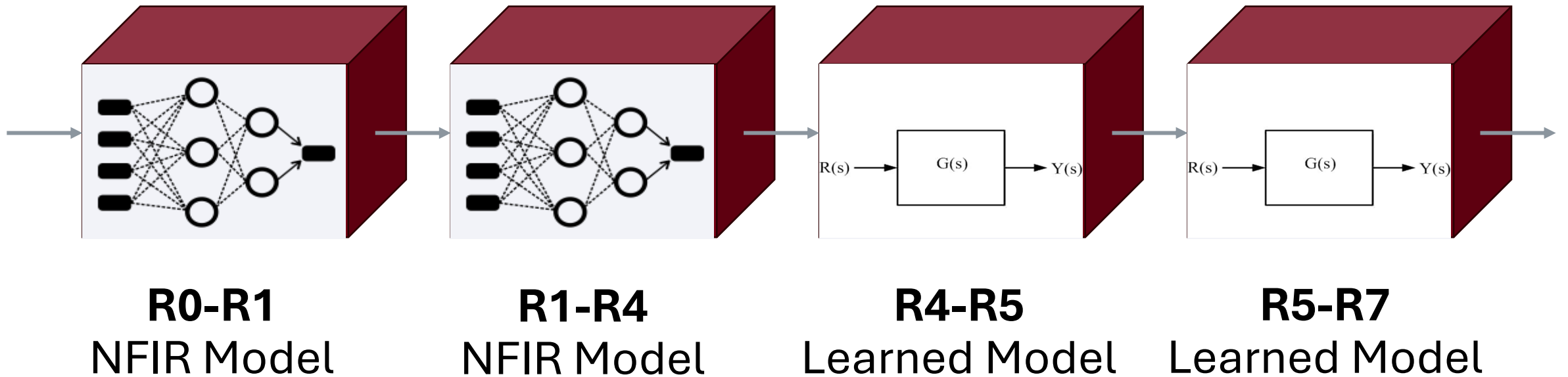
Training NFIR: NN-Based Modeling Approach

11 Different NFIR Models
With Multiple Permutations

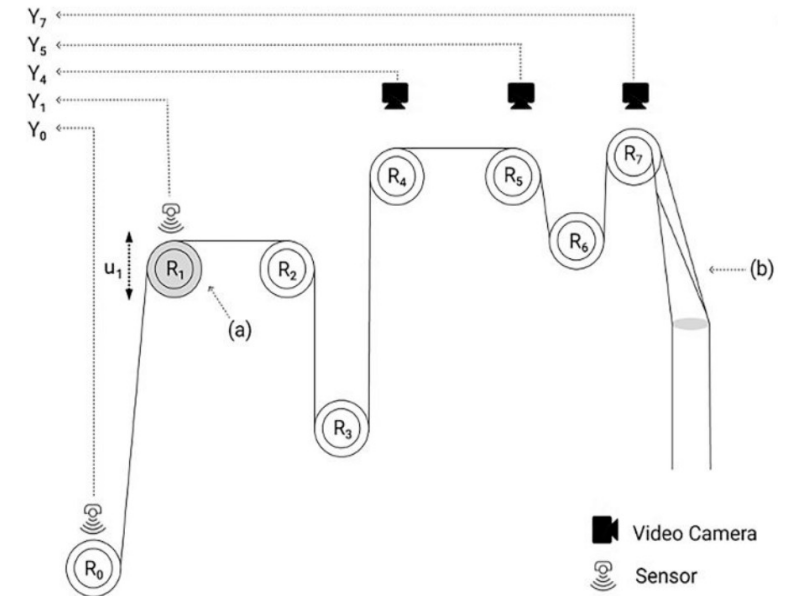
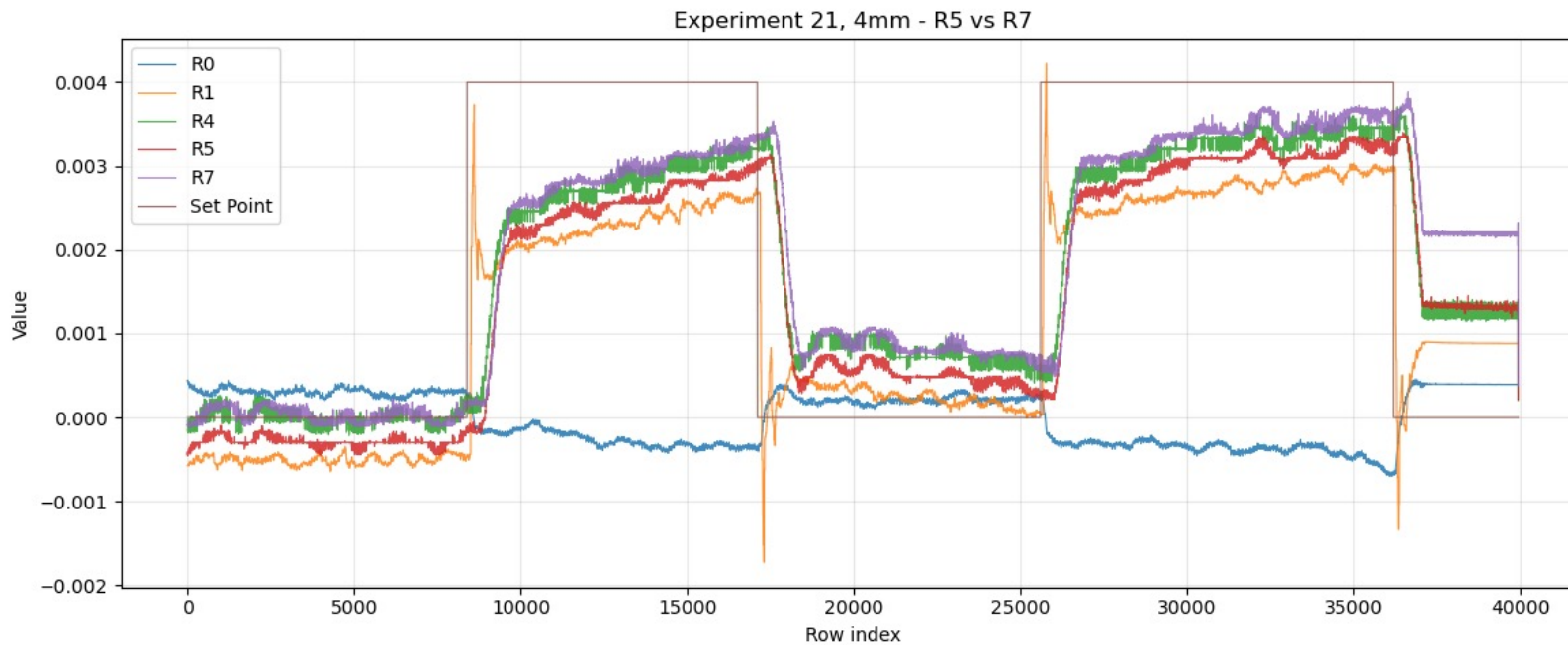


Complete Plant

Complete Plant

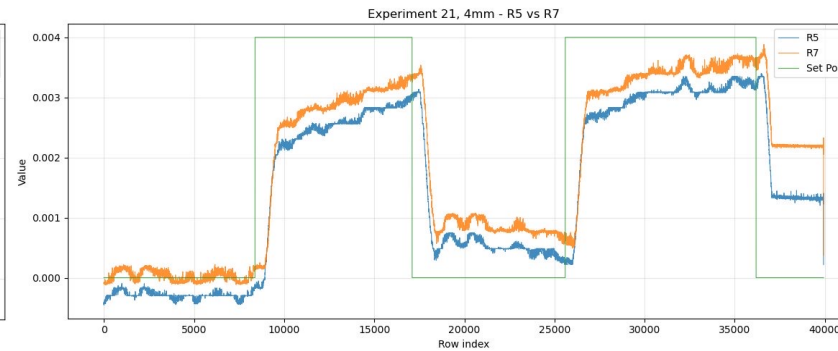
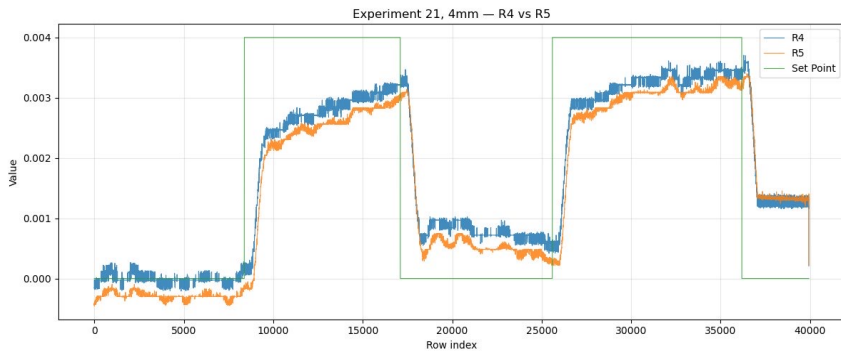
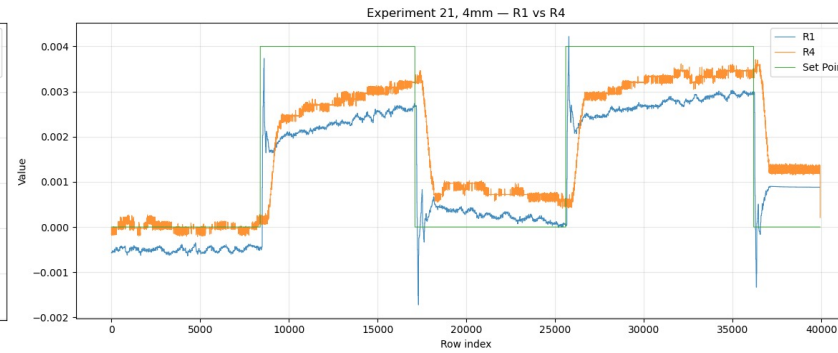
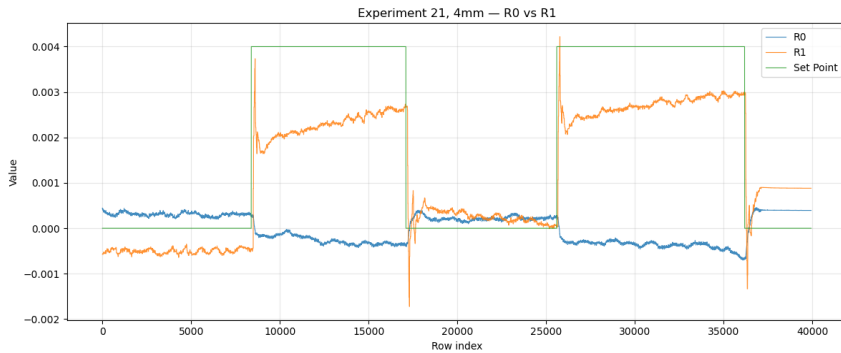


EDA – Different Inputs and Outputs

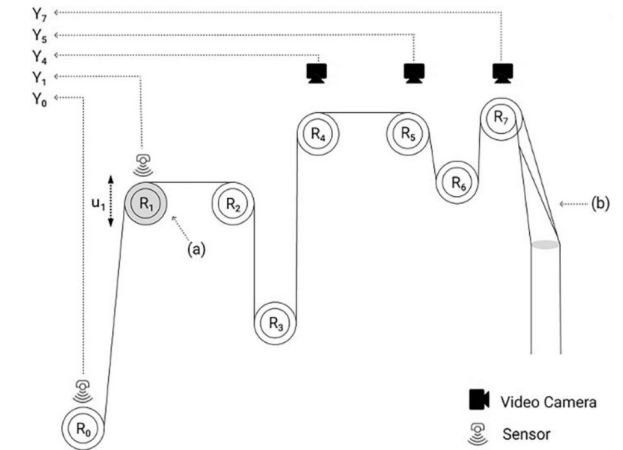


The plot R0, R1, R4, R5, and R7 corresponds to the position of the web at each roller. The setpoint corresponds to the position that U1 is intended to go to.

EDA – Different Inputs and Outputs



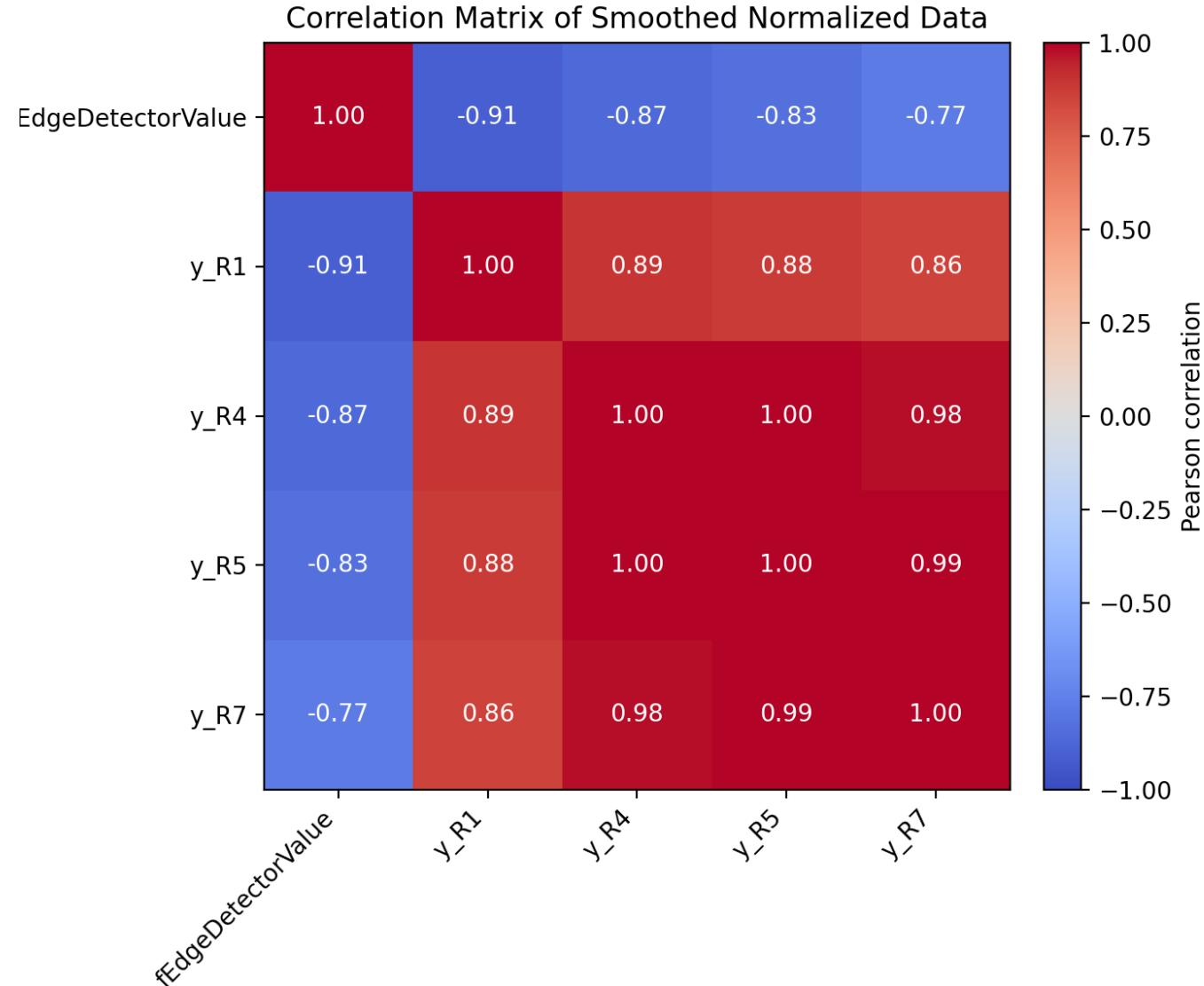
All of these instances shows that at different input roller, the output excitations shows a different model format. Therefore, it is useful to model individual roller to roller to reduce complexity and variability.



KEY TAKEAWAYS

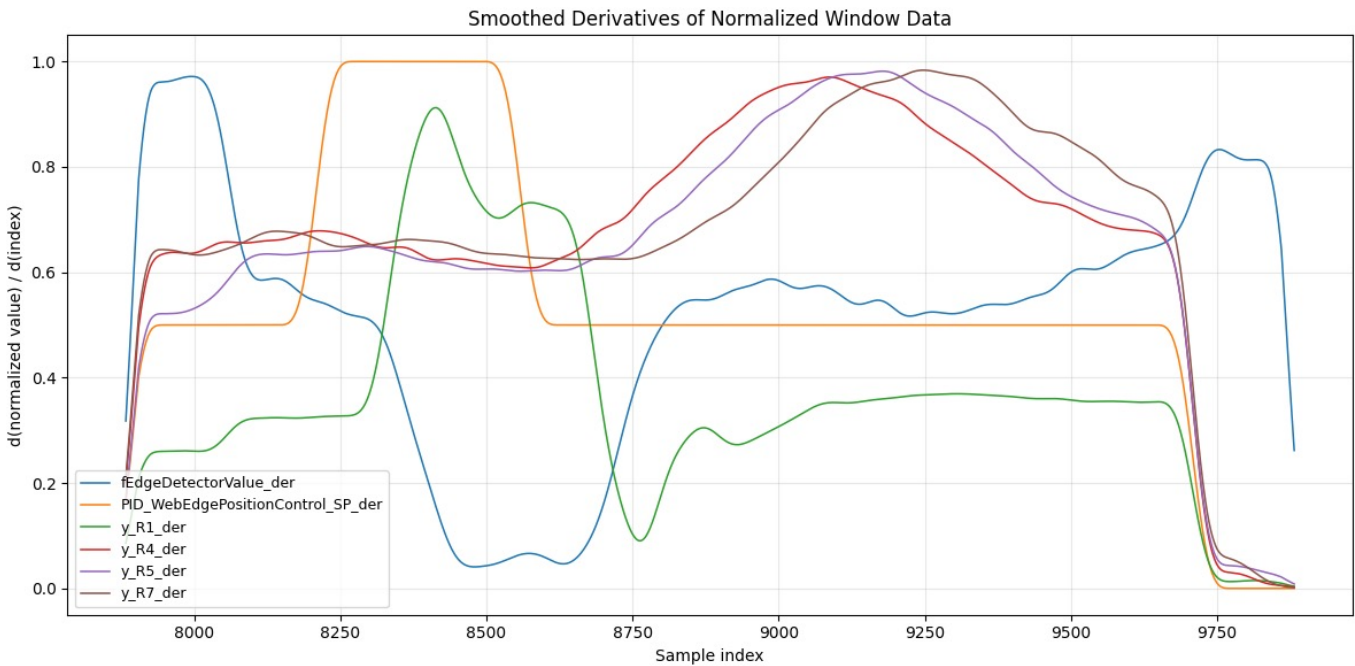
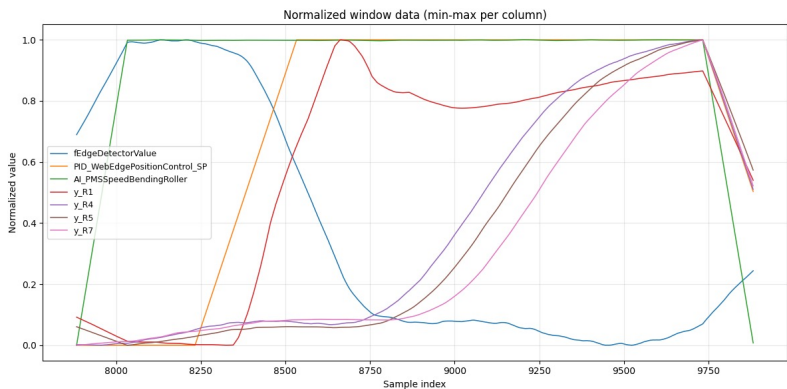
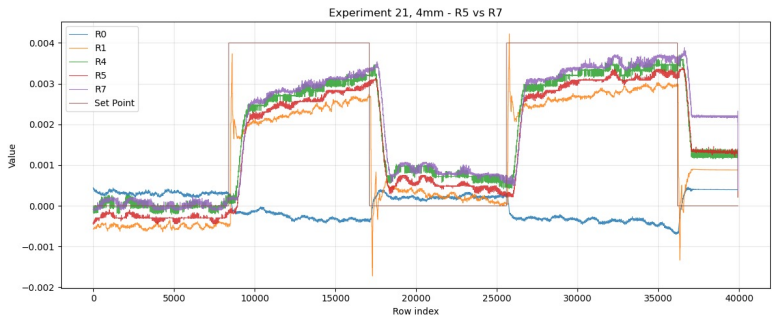
- **Spikes on R1** shows sharp transient spikes
- **R4,R5,R7** rollers behave like low-pass filters
- System response changes by roller location
- Supports **roller-to-roller modeling**

EDA – Correlation Matrix



- Low correlations may indicate nonlinear behavior or unexplained uncertainty (**epistemic uncertainty**).
- High correlations suggest strong linear relationships with remaining random variation (**aleatoric uncertainty**)

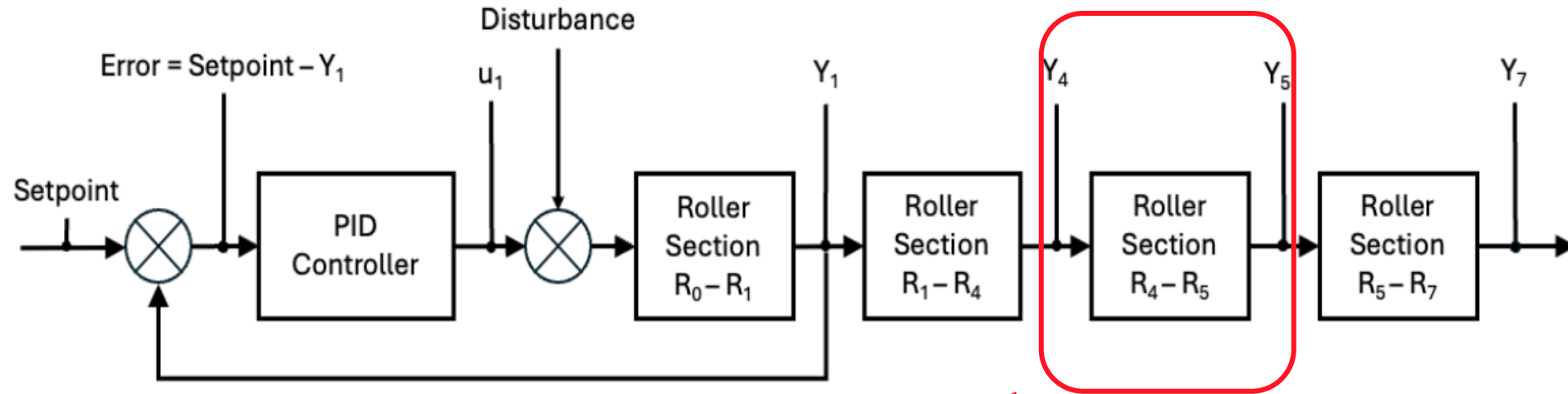
EDA - Delays



	R0	R1	R4	R5	R7
Delay from setpoint (Sec)	0.32	0	3.05	3.63	4.94

Delays will tell us the lag time required to understand how long it would take for the output would take effect. When we use the model to determine what the control loop would do, it would take into account the lag time between the input and output by its lag time.

Modeling Section R4-R5: Transfer Function

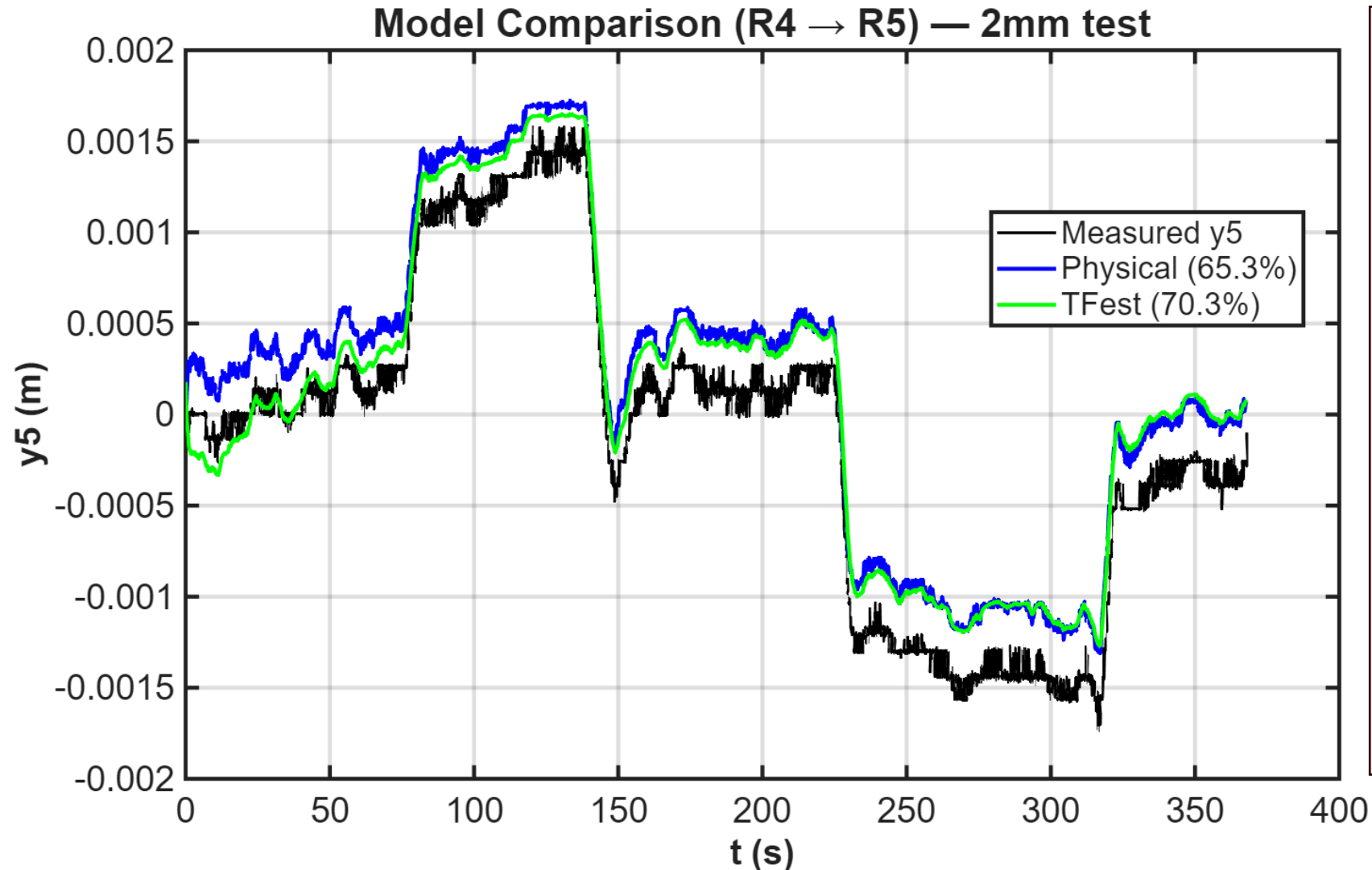


$$G_{4 \rightarrow 5}(s) = \frac{-\frac{f_3}{\tau} s + \frac{f_1}{\tau^2}}{s^2 + \frac{f_2}{\tau} s + \frac{f_1}{\tau^2}}$$

Where:

- s^2 : inertia/dynamic motion effects
- $\frac{f_2}{\tau} s$: Damping and stability behavior
- $\frac{f_1}{\tau^2}$: System stiffness and response speed

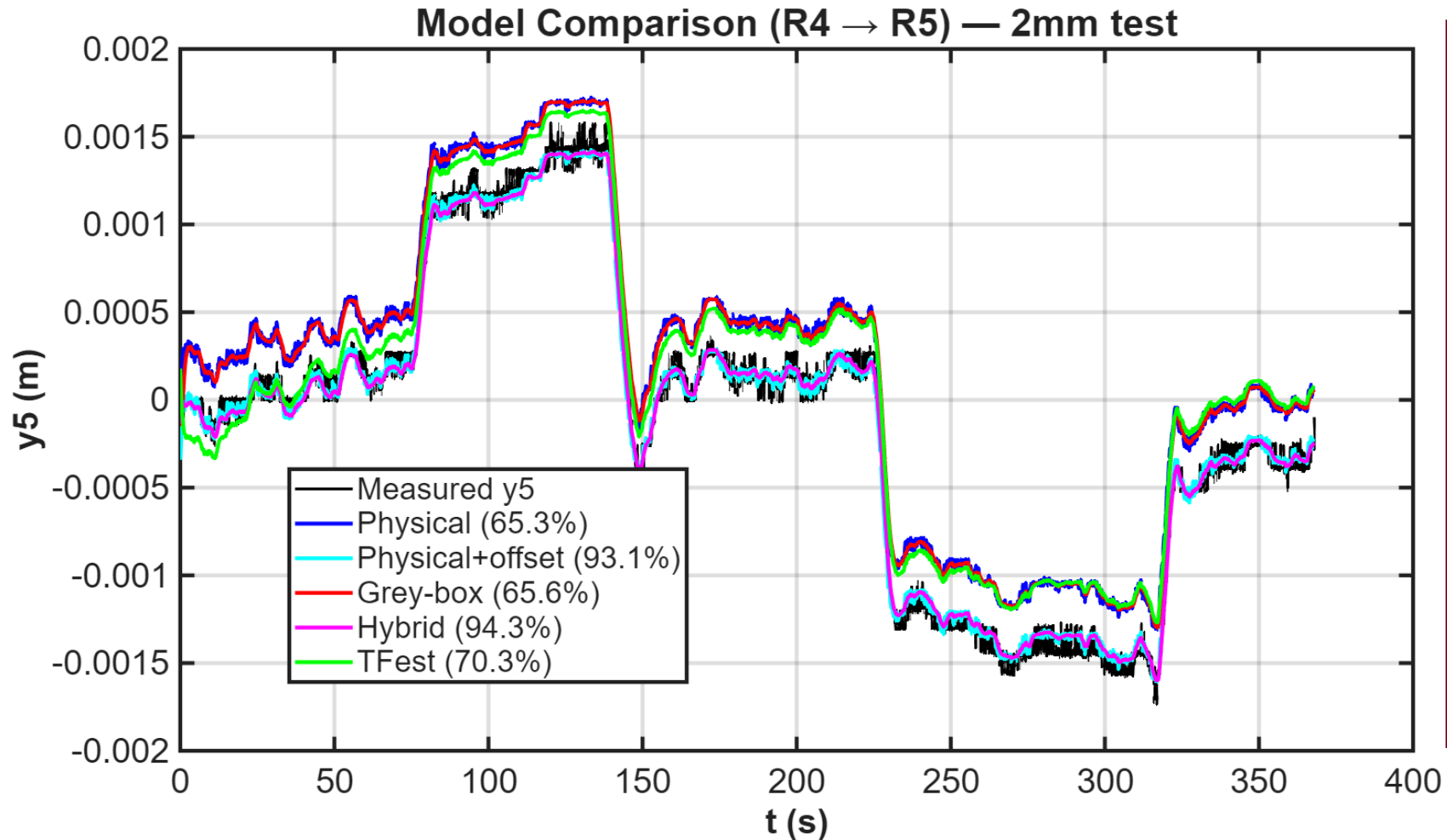
Modeling R4-R5: Initial Attempts



Physical Analysis

- **Physical model underperforms:**
 - Captures the overall dynamics, but fit accuracy remains low.
- **Transfer function improves performance:**
 - The black-box model achieves a better fit (70.3%) than the first-principles model.
- **Consistent output offset observed:**
 - Both models show a noticeable gap from the measured response.

Modeling R4-R5: Making Improvements

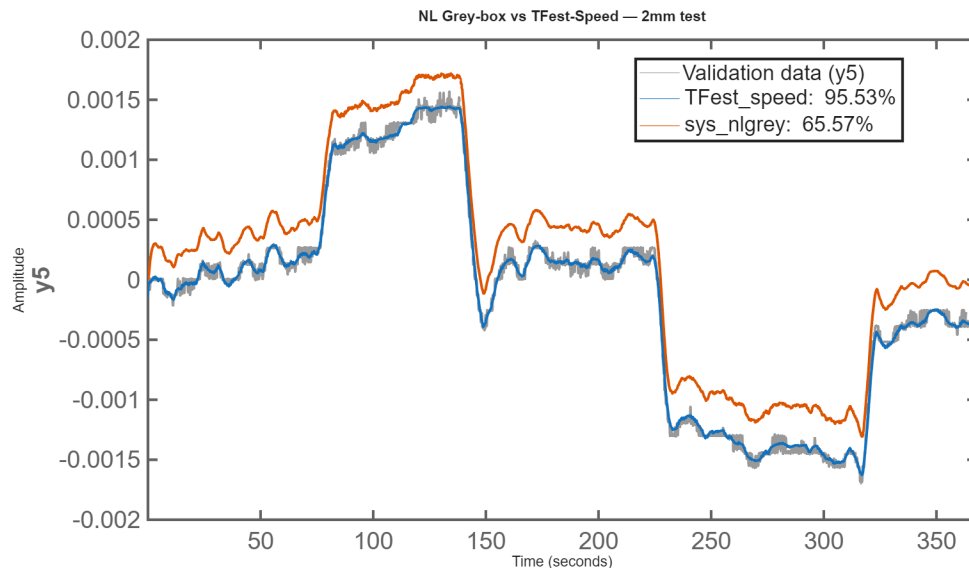


Physical Analysis

- **Grey-box model showed minimal improvement:**
 - Parameter optimization alone did not significantly improve fit.
- **Residual fitting greatly improved accuracy:**
 - Modeling the remaining error with a simple transfer function increased performance.
- **Adding a small DC offset closed the gap:**
 - Offset correction significantly improved agreement with measured data while preserving physical meaning

Modeling R4-R5: Making Improvements

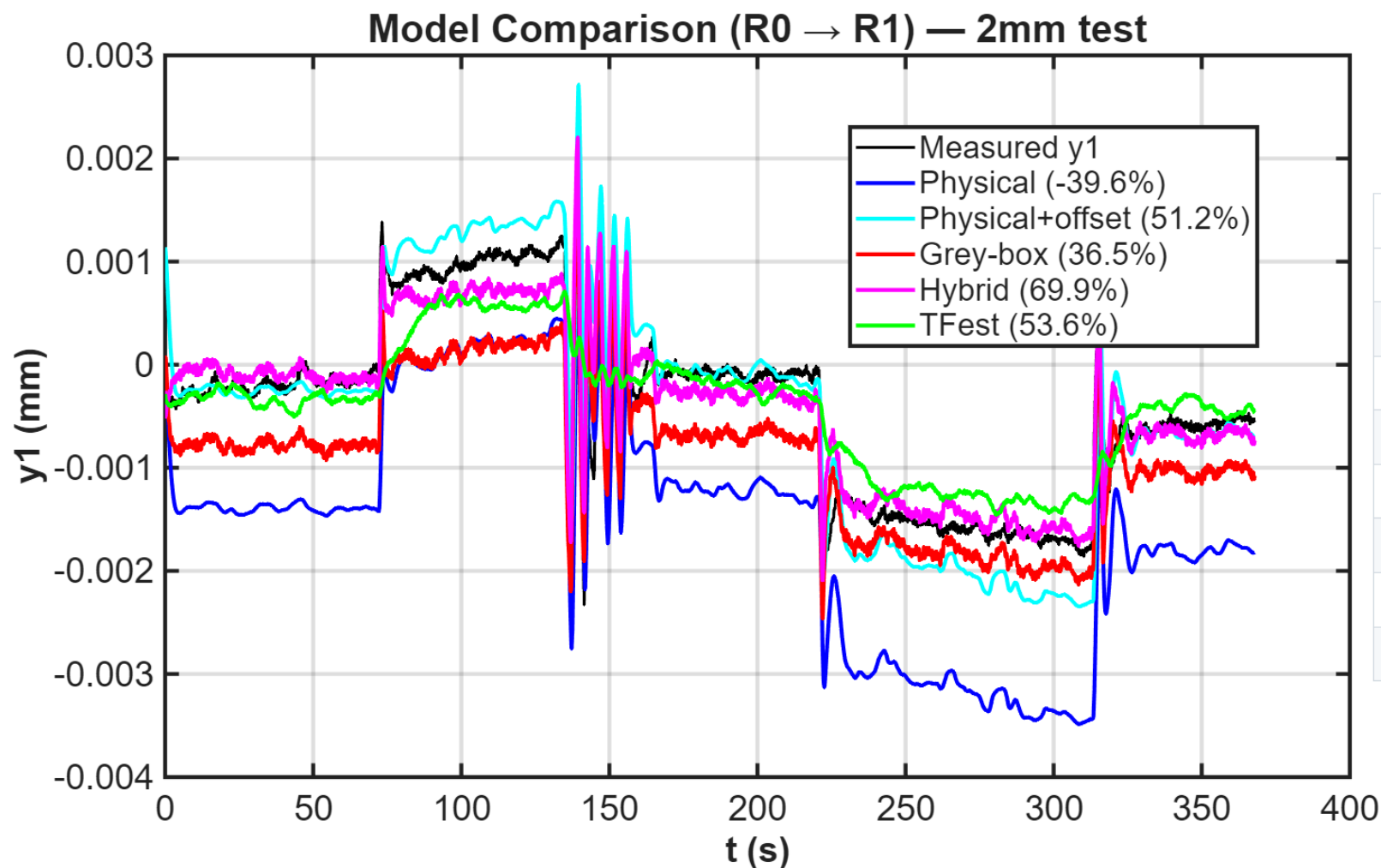
- Models with DC offset compensation significantly improved fit accuracy.
- The physical + DC offset model achieved strong performance while maintaining physical interpretability.



Model	4mm test	2mm test
Physical (beam equations)	89.0%	65.3%
Physical + DC offset	96.3%	93.1%
Black-box tfest (Y5 only)	89.6%	70.2%
Grey-box (greyest)	89.2%	65.6%
Hybrid (grey-box + residual)	96.3%	94.3%
Black-box tfest (Y4 + velocity)	97.7%	95.5%
Nonlinear grey-box (nlgreyest)	89.2%	65.6%
Black-box tfest (Y4 + velocity + U1)	96.5%	94.7%

Expanding Models: R0-R1

Very complex dynamics due to the moving U1 input

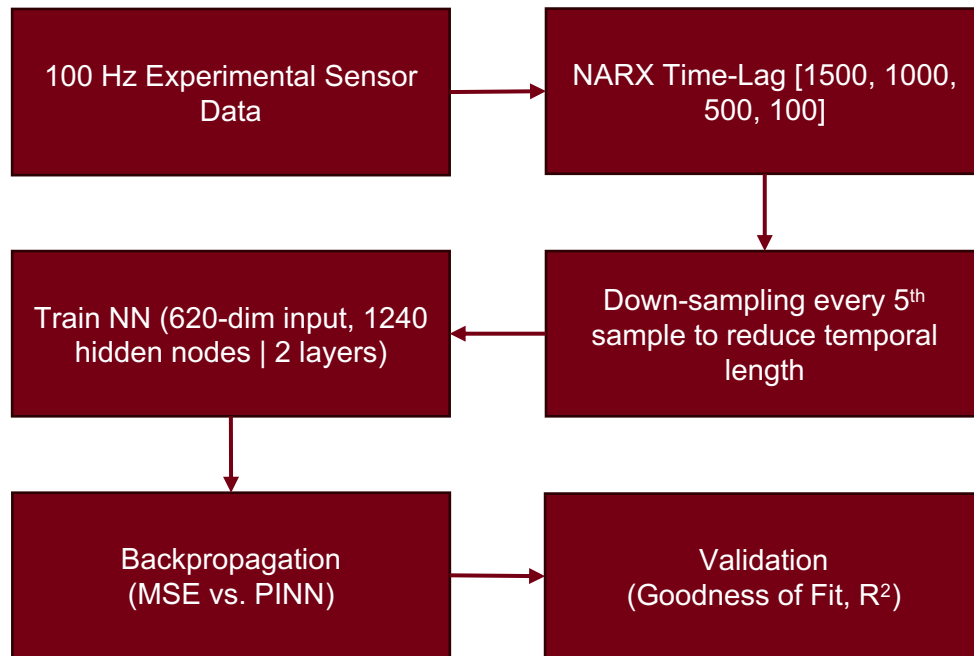


Model	4mm test	2mm test
Physical (beam equations)	39.5%	-39.6%
Physical + DC offset	66.2%	51.2%
Black-box tfest (Y0 + U1)	61.7%	53.6%
Grey-box (greyest)	65.5%	36.5%
Hybrid (grey-box + residual)	78.6%	69.9%
Black-box tfest (Y0 + U1 + velocity)	54.3%	52.5%
Nonlinear grey-box (nlgreyest)	N/A	N/A
Black-box tfest (Y0 + U1 + velocity + extra)	N/A	N/A

Neural Network Modeling of R0-R1 System

Complex Dynamics in R0-R1 produced poor fit of roughly 60-70% - Can we use PINN's to improve the model?

➔ Building a Neural Network



NN Tradeoff

Deep network with capability of capturing nonlinear dynamics, but **limited interpretability**.

🔧 Two Different Modeling Approaches

MSE-Based NN

Purely **data-driven**, tries to minimize error between predicted and measured R7 displacement

Physics Informed

Adds a **physics-based constraint**. Penalized predictions that violate the Sievers ODE linking R0 to R1 dynamics

Loss Calculations:

$$Loss = \underbrace{\frac{1}{N} \sum (\hat{y}^1 - y^1)^2}_{\text{Standard MSE}} + \lambda_{physics} \underbrace{L_{physics, Sievers ODE}}_{\text{Mean Squared Residual of Sievers ODE}}$$

Standard MSE

Mean Squared Residual of Sievers ODE

R0 → R1 Modeling Setup & Results

NON-SPLICE MODEL

310,593

Downsampled Training Samples

Training (6 Runs)

21_4mm_test2	22_2mm_fast
24_step_gain_1_75	25_step_gain_2_6
27_4mm_fast	30_2mm

Testing (2 Runs)

28_4mm	29_2mm_test2
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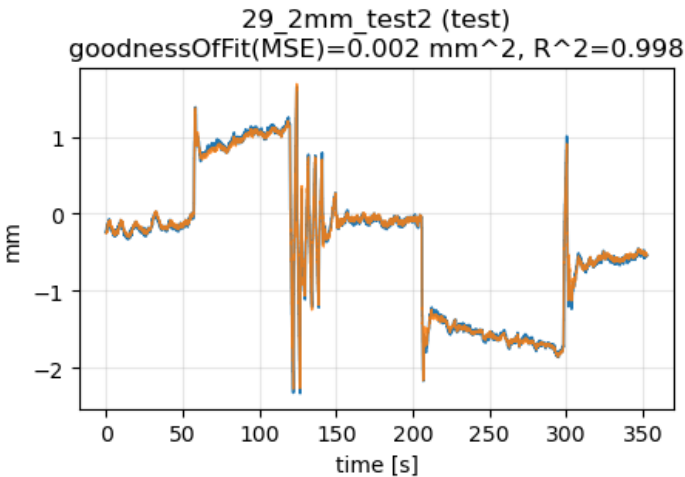
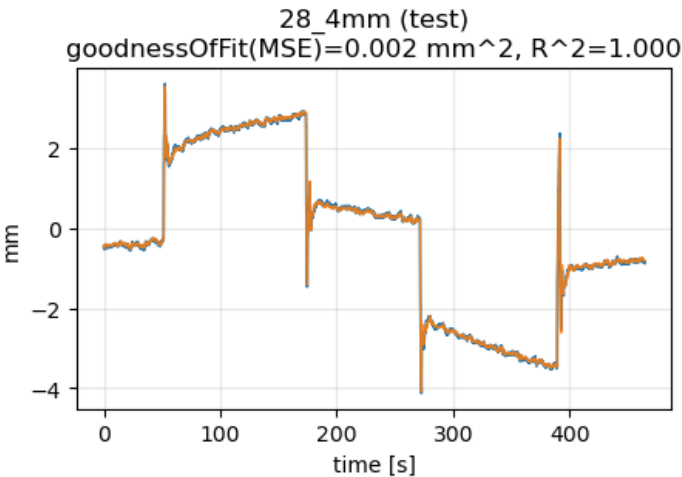
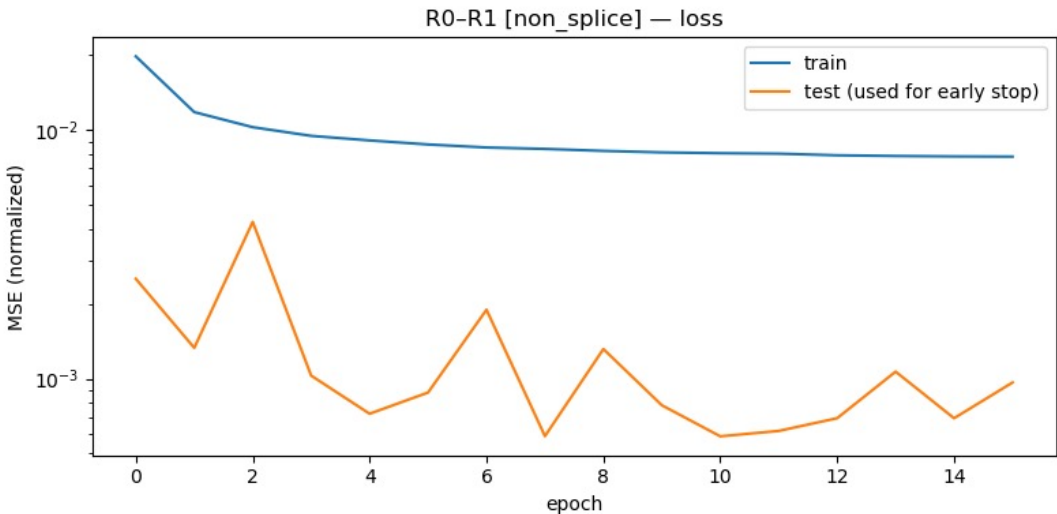
DOWNSAMPLED
Temporal compression
applied to all signals.



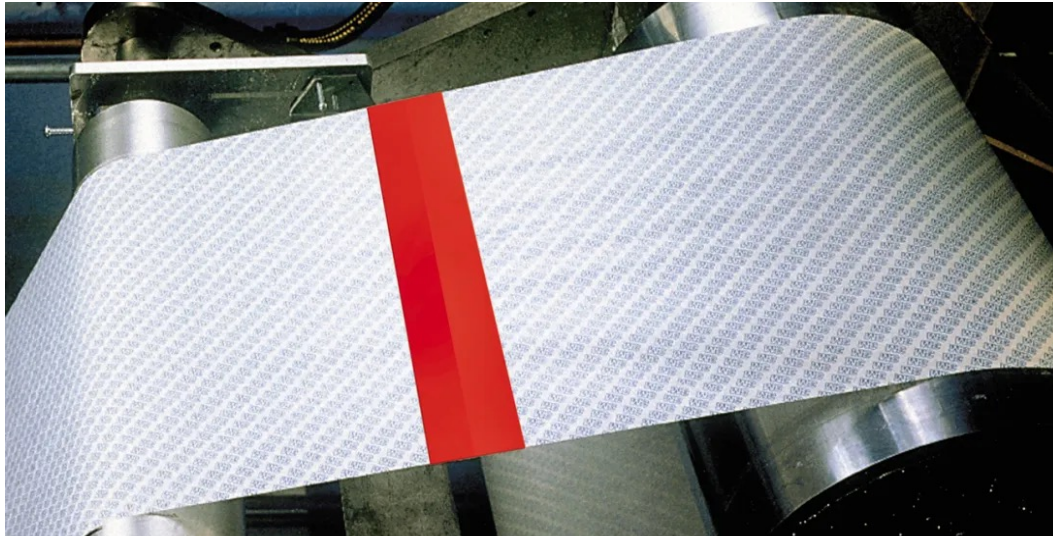
FILM CONDITIONS
Uniform environmental
states across runs.



DYNAMICS
Variation: 2mm vs 4mm
roller positions.



Splice vs. Non-Splice Operating Regimes

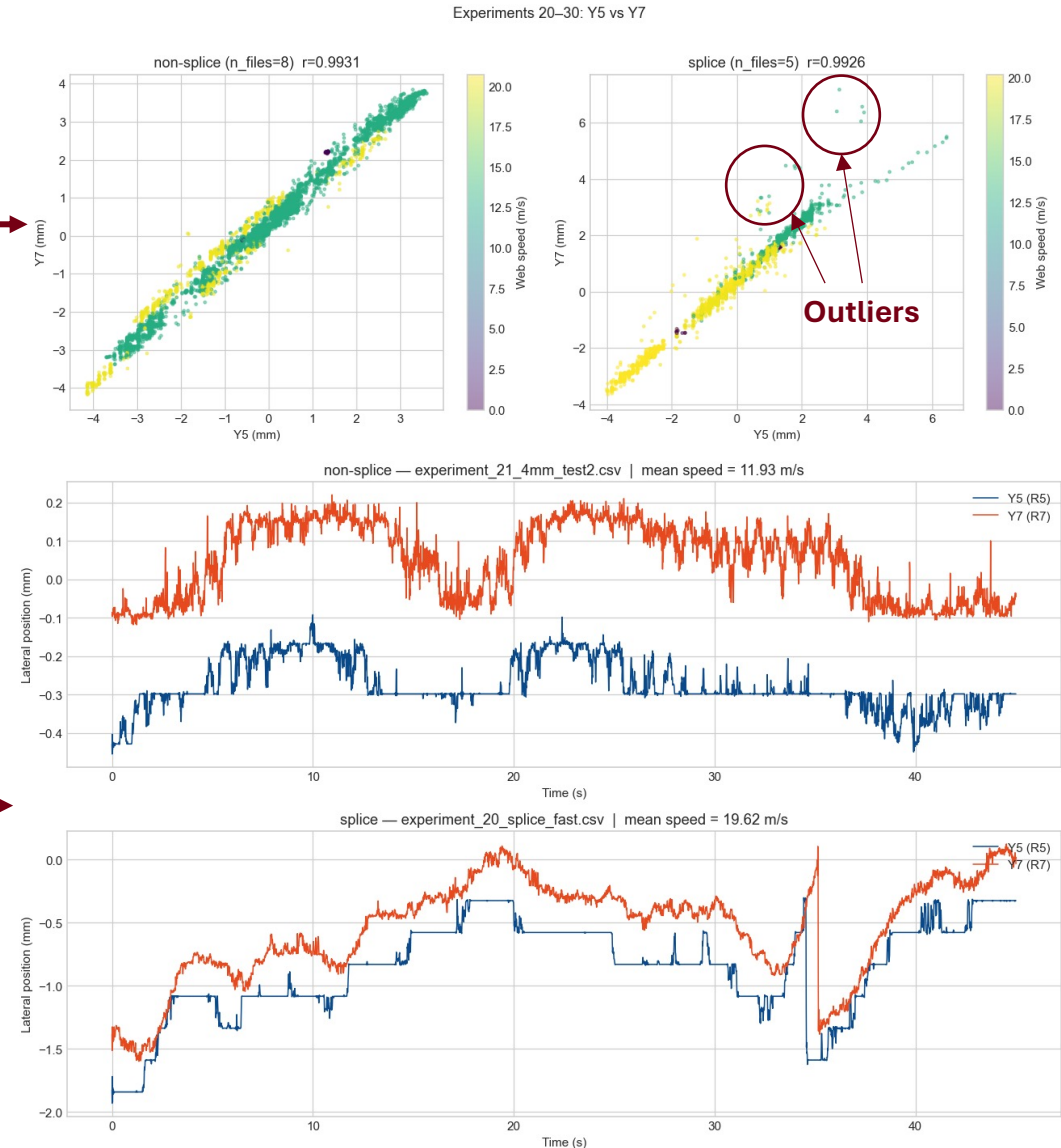


Non-Splice Regime

Steady-state operation where mass distribution remains uniform across the web path.

Splice Regime

A discontinuity point where two separate rolls are joined using adhesive.



MSE Based NN for Modeling R0-R1 Splice

 **SPlice MODEL**

272,970

Dynamic Event Modeling

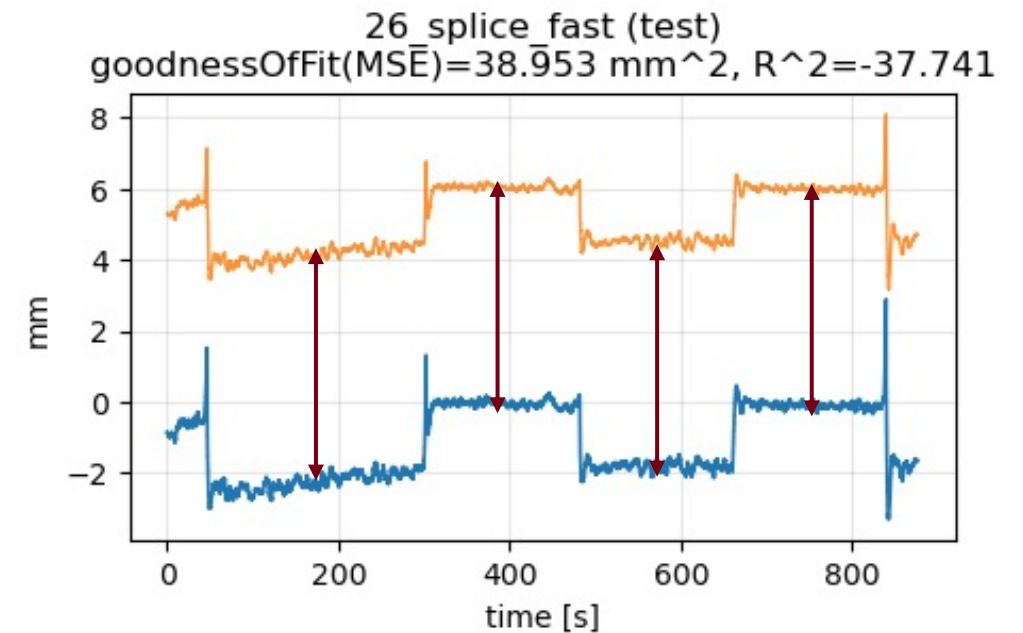
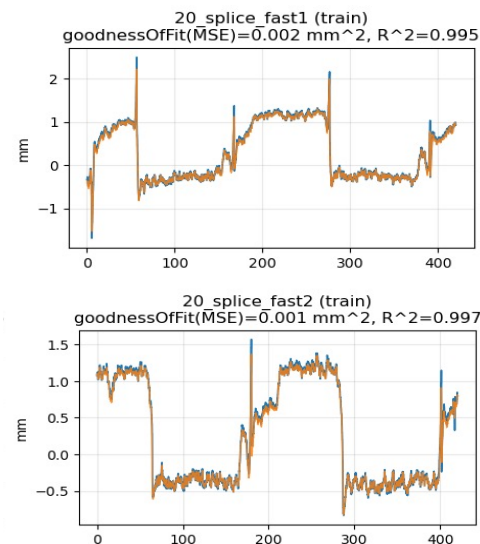
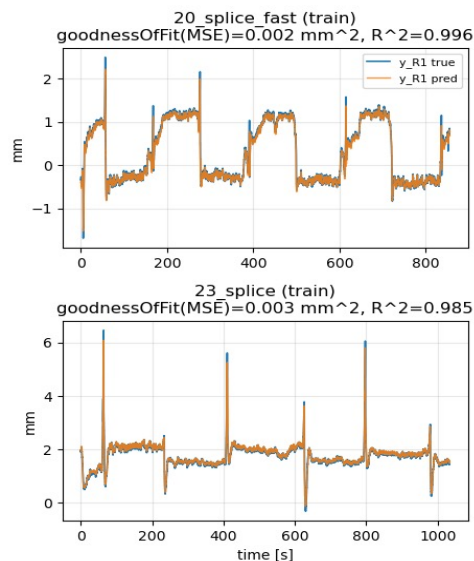
Training (4 Runs)

3x experiment_20_splice

experiment_23_splice

Testing (1 Runs)

experiment_26_splice



PINN for Modeling R0-R1 Splice

LOSS CALCULATIONS

$$Loss = \underbrace{\frac{1}{N} \sum (\hat{y}^1 - y^1)^2}_{\text{Standard MSE}} + \underbrace{\lambda_{physics} L_{physics, Sievers ODE}}_{\text{Mean Squared Residual of Sievers ODE}}$$

PHYSICS-BASED MODEL

Using Siever's 2nd-order ODE physics model to define baseline web dynamics (Physics Explained in Prior Slides)

 f_1

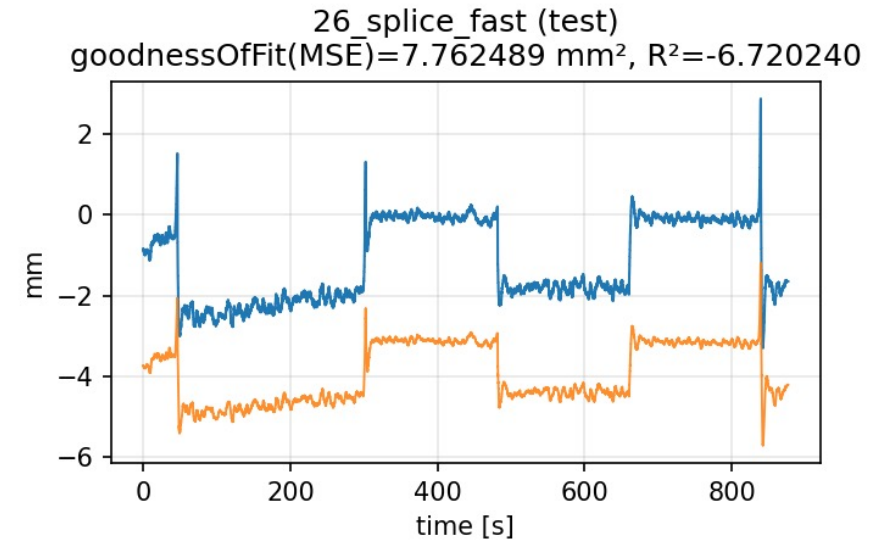
Natural Frequency
Position Feedback

 f_2

Damping
Velocity Effects

 f_3

Material Properties
R0 + Actuator → R1



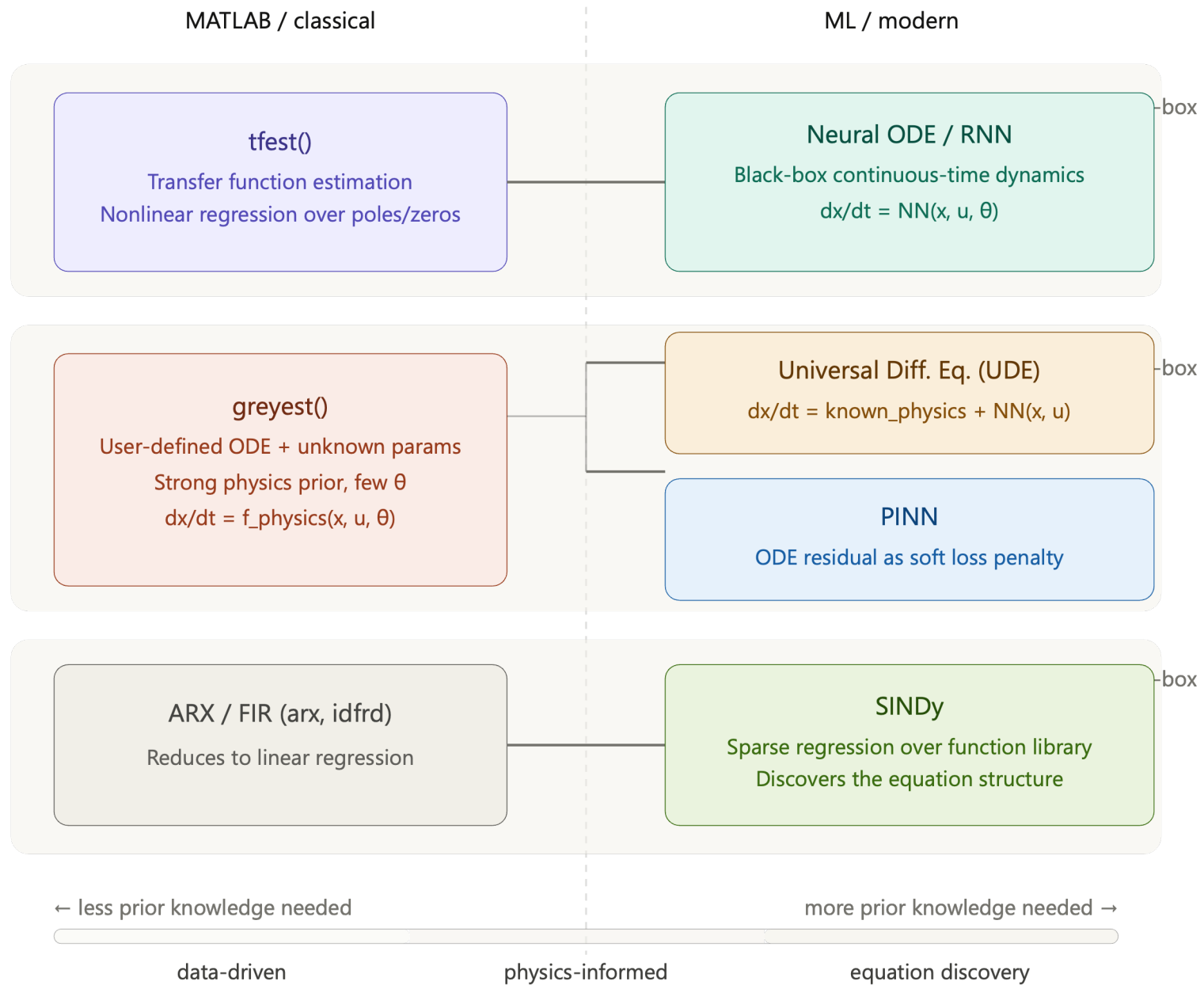
- **Physics mainly constrains shape, not offset** → learns how the signal changes over time (slopes and curvature), but it does not strongly fix the offset
- **Limited data diversity per regime** → training may not fully span steady-state variations seen in unseen experiments
- **Future work (mixed-regime training)** → train on a combination of splice + non-splice data and evaluate on both to test cross-regime generalization and bias robustness

Conclusion

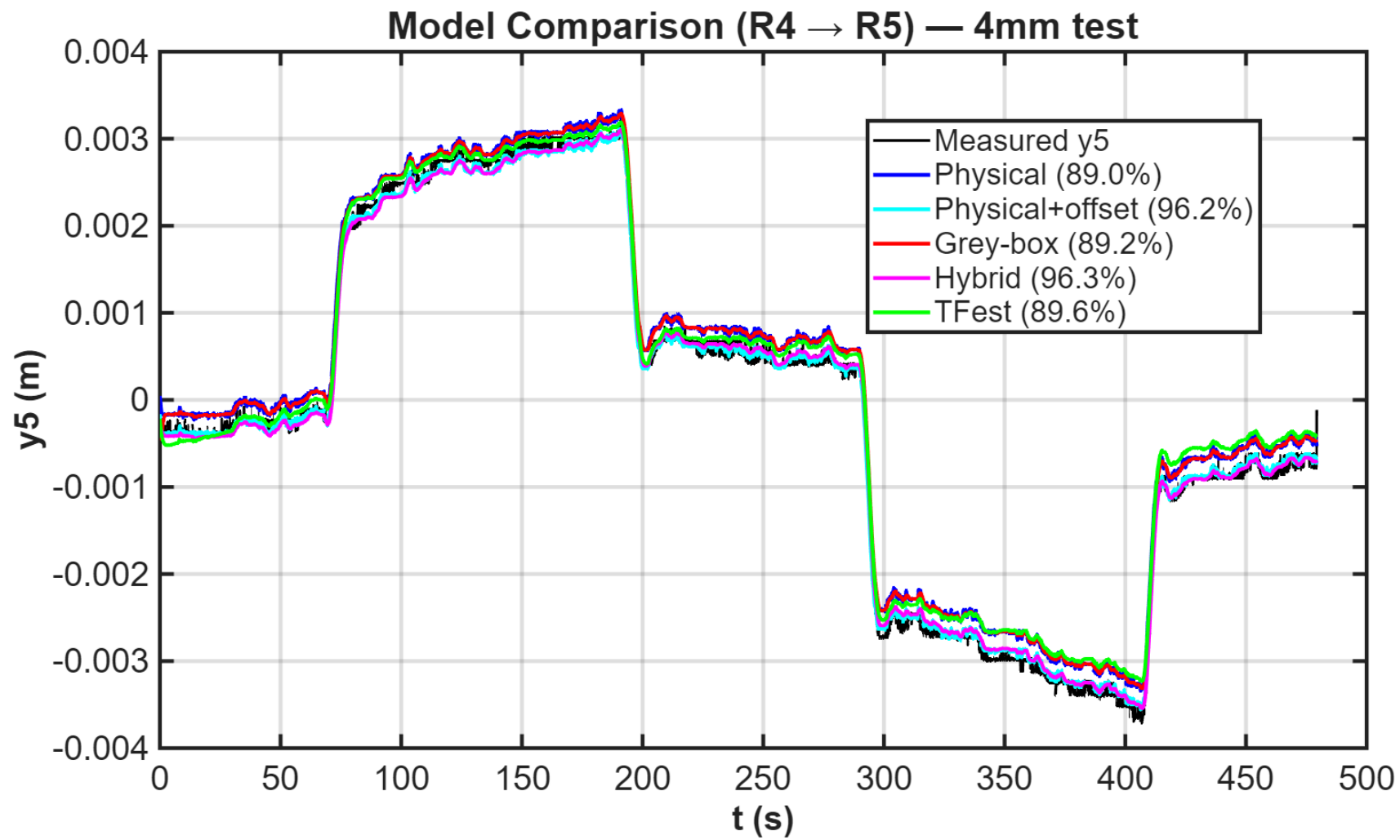
- 01 Roll-to-roll web dynamics vary significantly between roller sections, motivating a modular roller-to-roller modeling approach.
- 02 Neural network-based NFIR/PINN approaches helped model highly nonlinear and splice-driven dynamics in the R0–R1 region.
- 03 Physics-informed transfer functions captured overall system behavior, but required DC offset compensation to achieve high accuracy.
- 04 Future work includes controller optimization, real-time deployment, and improved modeling of transient splice events.

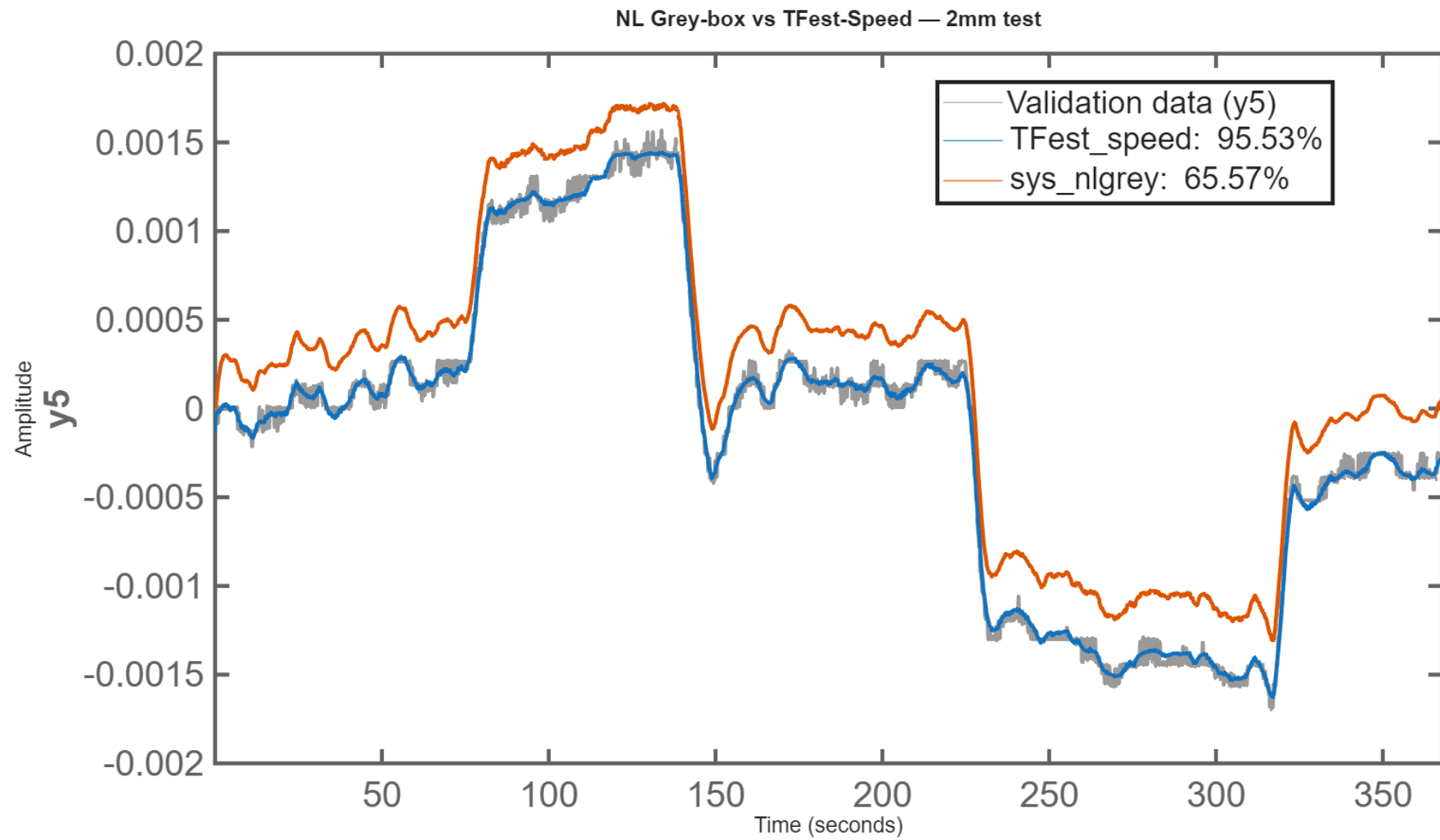
Combining physics-informed and data-driven modeling provides a scalable framework for improving control, reducing waste, and increasing robustness in roll-to-roll manufacturing systems.

Backup



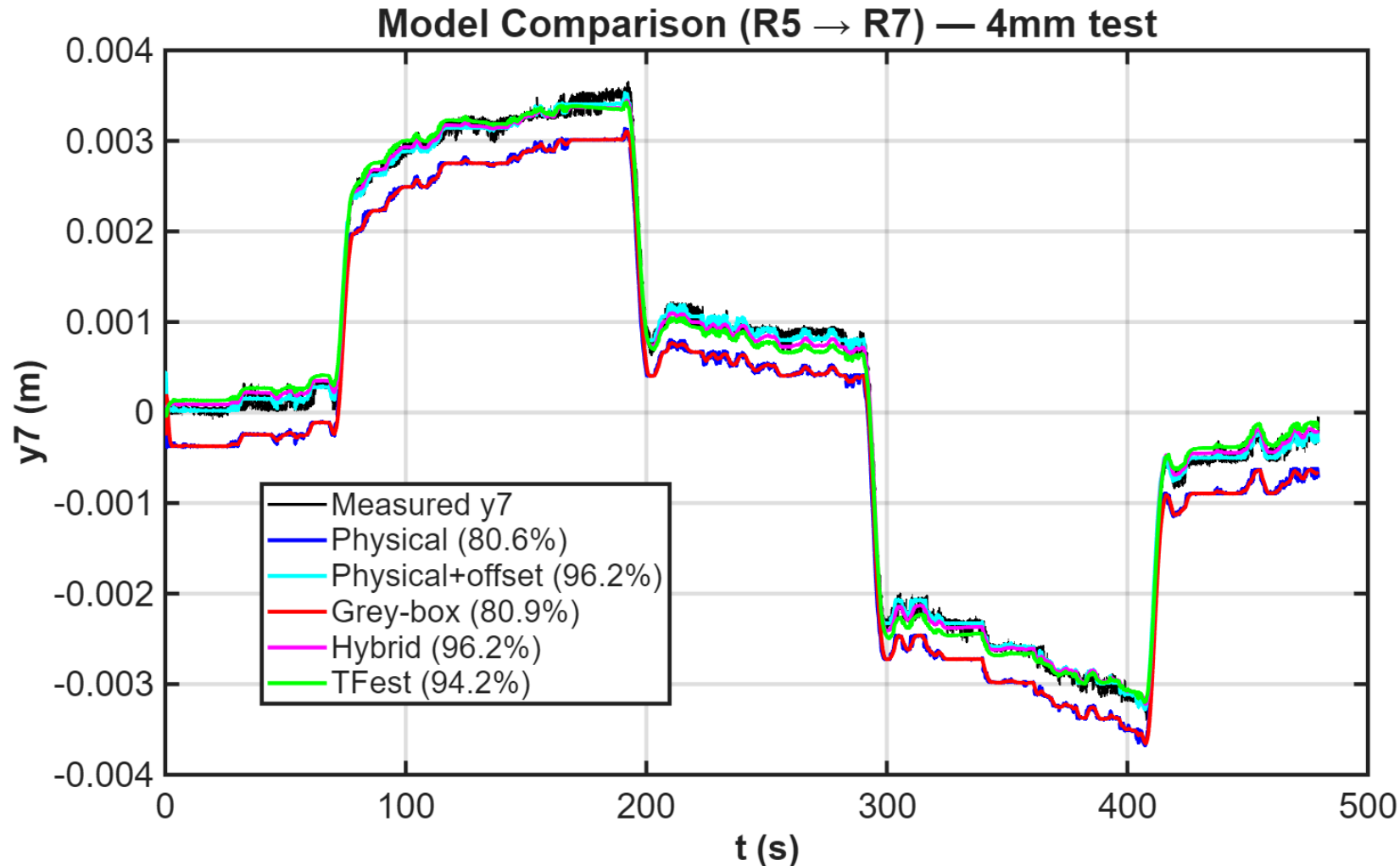
Model	R4-R5 / R5-R7	R1-R4	R0-R1
Physical	Correct dynamics, wrong DC reference	Genuinely wrong dynamics — three compounding sources of error	Most severely wrong — longest span, idealized actuator boundary, closed-loop operation
Physical + DC offset	Near-ceiling (~96%)	Insufficient (~68% avg) — errors are dynamic	Poor (~59% avg) — dominant dynamic errors dwarf the bias
Black-box tfest	Reasonable; absorbs DC offset implicitly for R5-R7	Catastrophic overfitting from correlated inputs	Mediocre but not catastrophic — y_0 and u_1 are less correlated than y_1 and u_1
Grey-box	Same DC reference error as physical	Protective against overfitting despite poor physical model	Can overfit amplitude imbalance in training — worse than physical+offset for 2mm
Hybrid	DC offset absorbed by residual; near-ceiling	Residual captures genuine dynamic errors; 56 pp improvement for 2mm	Best available model; still only ~74% avg — linear family structurally insufficient
tfest with velocity	Per-experiment bias lookup; large improvement	Same mechanism	Slightly harmful — velocity channel cannot cleanly absorb bias with competing low-frequency inputs
tfest with velocity + U_1	Severe overfitting (R5-R7); modest drop (R4-R5)	Not applicable	Not applicable





Expanding Models: R5-R7

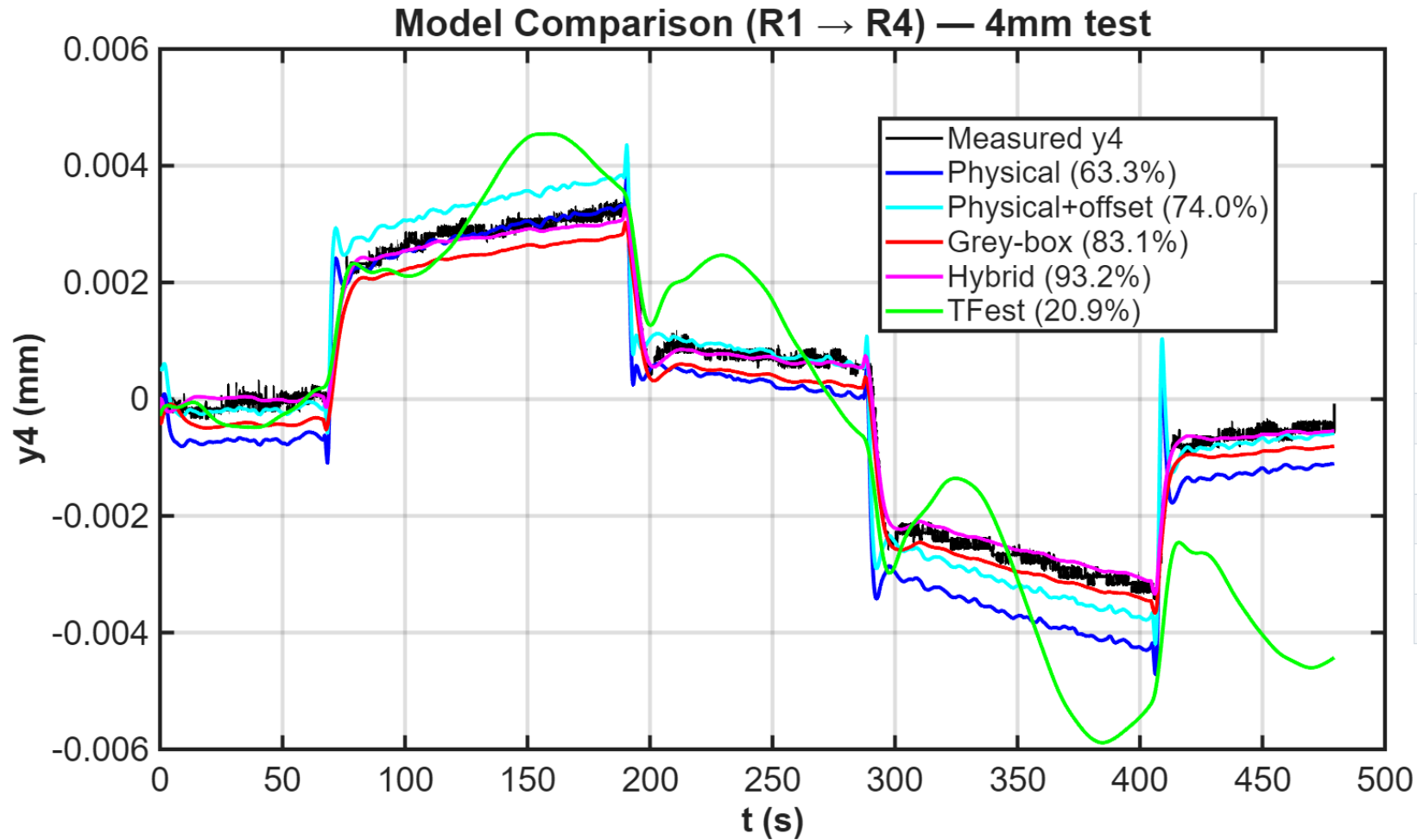
R5-R7 provides a slightly more complex system to model than R4-R5, with 2 rollers instead of 1



Model	4mm test	2mm test
Physical (beam equations)	80.6%	77.3%
Physical + DC offset	96.2%	92.1%
Black-box tfest (Y5 only)	94.2%	93.0%
Grey-box (greyest)	80.9%	78.0%
Hybrid (grey-box + residual)	96.2%	93.7%
Black-box tfest (Y5 + velocity)	97.3%	94.5%
Nonlinear grey-box (nlgreyest)	N/A	N/A
Black-box tfest (Y5 + velocity + U1)	-49.4%	-31.9%

Expanding Models: R1-R4

R1-R4 needs to model 3 rollers, as well as a position R1 that is moving with the input U1



Model	4mm test	2mm test
Physical (beam equations)	63.3%	-3.4%
Physical + DC offset	74.0%	61.6%
Black-box tfest (Y1 + U1)	20.9%	-57.7%
Grey-box (greyest)	83.1%	34.5%
Hybrid (grey-box + residual)	93.2%	90.6%
Black-box tfest (Y1 + U1 + velocity)	94.1%	89.2%
Nonlinear grey-box (nlgreyest)	N/A	N/A
Black-box tfest (Y1 + U1 + velocity + extra U1)	N/A	N/A

R0-R1: Using a NN/NARX-based approach

- A NARX, or Nonlinear Autoregressive Exogeneous, is a nonlinear Neural Network-based approach to modeling time-series data based on purely data-driven methods.
 - Although this may provide higher metrics, it loses its physical interpretability entirely, meaning it may be unsuitable under altered conditions and is learning complex patterns rather than the true underlying physics.
- **The main point:** *There's a fine balance between model performance and model interpretability.*

